Link Budget Analysis and Simulation of Free-Space Optical Communication Systems for Deep Space

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ABSTRACT

Free-space optical communication can provide data rates to deep-space missions that radio links cannot. Working out whether a given link will close means calculating each gain and loss in turn, and the tools that do this for the missions that have flown are not public. This dissertation presents an open Python simulator for free-space optical satellite links. It starts from a first-principles link budget and is extended in stages to include atmospheric effects, photon-counting pulse-position modulation, channel coding and coherent detection. The simulator is checked by 65 automated tests, 48 reproductions of published calculations and 17 comparisons with measurement: published link budgets are reproduced to within 0.1 dB, and the GOPEX laser uplink experiment is matched to within 0.6 dB of its own theory, the modelled DSOC decode threshold lies 0.25–1.6 dB below the mission's published operating point across its background range, and the modelled turbulence fading lies inside the spread measured on 40 cm ground-station downlinks at 1550 nm. Of the flown missions benchmarked, only DSOC was able to test the rate model; LLCD's flown rate is set by its modem's slot clock, so it can only confirm that the link closes. The simulator is then applied to several design questions. A sensitivity analysis shows that the telescope apertures matter more than any other link parameter. A study of coherent modulation with a stabilised laser gives the linewidth required at deep-space data rates, and a component-level CubeSat terminal is sized at 550 g and 8.5 W against published mass and power budgets. Finally, the University of Surrey's own 40 cm telescope, fitted with an assumed photon-counting receiver, could receive about 129 kb/s (32–275 kb/s across the assumed values) from a DSOC-class transmitter at 2.68 AU on a clear night, provided the signalling is optimised for the weaker link.

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Caption prefixes: *Simulator output* marks values computed by the simulator of Chapter 3; *Published values* marks values taken from the cited sources; *Validation* marks simulator output compared with published values. Descriptive tables carry no prefix.

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LIST OF ABBREVIATIONS

APD	Avalanche Photodiode
AU	Astronomical Unit
BER	Bit Error Rate
BPSK	Binary Phase-Shift Keying
CCSDS	Consultative Committee for Space Data Systems
DFB	Distributed Feedback (laser)
DLR	Deutsches Zentrum für Luft- und Raumfahrt (German Aerospace Center)
DPSK	Differential Phase-Shift Keying
DSOC	Deep Space Optical Communications
EDFA	Erbium-Doped Fibre Amplifier
EDRS	European Data Relay System
FEC	Forward Error Correction
FPGA	Field-Programmable Gate Array
FSO	Free-Space Optical (communication)
GEO	Geostationary Earth Orbit
GOPEX	Galileo Optical Experiment
HV	Hufnagel–Valley (turbulence profile)
ISL	Intersatellite Link
ISS	International Space Station
ITU-R	International Telecommunication Union, Radiocommunication Sector
JPL	Jet Propulsion Laboratory
KIODO	Kirari Optical Downlink to Oberpfaffenhofen
LADEE	Lunar Atmosphere and Dust Environment Explorer
LCT	Laser Communication Terminal
LDPC	Low-Density Parity-Check (code)
LEO	Low Earth Orbit
LLCD	Lunar Laser Communication Demonstration
MODTRAN	Moderate-resolution Atmospheric Transmission (model)

NFIRE	Near Field Infrared Experiment
NODE	Nanosatellite Optical Downlink Experiment
NRLMSISE-00	Naval Research Laboratory Mass Spectrometer and Incoherent Scatter (atmosphere model, 2000 edition)
ODK16	Orion Optics Dall–Kirkham 16-inch telescope
OICETS	Optical Inter-orbit Communications Engineering Test Satellite
OOK	On-Off Keying
OPALS	Optical Payload for Lasercomm Science
PPM	Pulse-Position Modulation
RF	Radio Frequency
SCPPM	Serially-Concatenated Pulse-Position Modulation
SILEX	Semiconductor-laser Intersatellite Link Experiment
SNSPD	Superconducting Nanowire Single-Photon Detector
SOLT	Strategic Optical Link Tool (JPL)
TBIRD	TeraByte InfraRed Delivery
TT&C	Telemetry, Tracking and Command

1 INTRODUCTION

1.1 Background and Context

The value of a space mission comes from what can be learned from it, and that depends on the amount of data that can be returned to Earth. For over fifty years, radio has been the default carrier for satellite links. It propagates through cloud and rain with modest loss, its beamwidth is wide enough that pointing errors cost little, and it has decades of infrastructure already in place. However, with ever-growing demand for communications capacity worldwide, allocated bands are growing more congested and strictly regulated. New allocations are slow to license, and the achievable data rate now falls short of the volumes generated by modern instruments [2].

Free-space optical (FSO) communication offers a way around the growing congestion of RF links. The optical spectrum is not subject to the same allocation and licensing regime, and a beam from a 1550 nm laser and a small telescope is far narrower than a radio beam. DSOC's 22 cm flight telescope produces a 13.8 μrad beam that, from 2.68 AU, lights a spot about 5 500 km across at the Earth, smaller than the planet, where an X-band beam from a 2 m antenna spreads to some 13.5 million km (both as full $1/e^2$ widths) [3, 4], so interference on the ground is reduced. Although free-space path loss increases for optical links because of their shorter wavelength [5], transmit gain scales as $(\pi D/\lambda)^2$ [5, 6], so more of the transmitted power reaches the receiver. This is what makes optical links attractive for deep space. Distance costs radio and light the same 20 dB for every tenfold increase in range, but the gain of an aperture scales as $(\pi D/\lambda)^2$, so at 1550 nm a telescope of tens of centimetres matches a radio antenna of tens of metres [4]. Near the Earth that surplus is rarely needed; at Mars and beyond, however, it decides whether a link closes [7]. Table 1.1 shows the scale of the difference: for the same data rate from Mars, the optical terminal needs an aperture an order of magnitude smaller and a fraction of the transmit power.

These advantages have been demonstrated in real missions too; in 2013 LLCD returned 622 Mb/s from lunar orbit on the downlink and 20 Mb/s on the uplink [8], and DSOC has since sent 267 Mb/s from 0.37 AU, where radio links can provide only tens of Mb/s [3, 9].

The narrow beam that gives an optical link its gain is also its main drawback. DSOC's downlink beam is 13.8 μrad wide [3], and a beam that narrow must be held on the receiver to better than

Table 1.1: Published values: smallest spacecraft antenna and average transmit power needed to return each data rate from Mars at 2.67 AU, by Ka-band radio and by a 1.55 μm optical link, from [7] (Table 2).

	Ka-band radio		Optical, 1.55 μm	
Data rate (Mb/s)	Antenna (m)	Power (W)	Telescope (cm)	Power (W)
10	2.3	155	14	2
100	4.2	464	26	7
1000	9	1000	50	19

a microradian: an order of magnitude finer than its own width, and more than a thousand times finer than the 0.1° body pointing of a deep-space radio antenna [4, 10, 11]. Additionally, the atmosphere adds absorption and scattering, and cloud or daylight can interrupt the link, causing it to drop entirely. However, a link budget equation will be able to assess whether a link can close, by taking the gain or loss of each element in the chain and calculating the power arriving at the receiver. The signal remaining at the end is compared with what the receiver needs, which shows directly whether a link will close.

The theory behind each element of a deep-space link budget is well documented and published, but the open-source tools that pull them together and form them into a complete link budget are not. For example, JPL's SOLT link budget tool, used by NASA, is described in the literature but is not publicly accessible [12]. No openly published deep-space optical link budget tool was found, and published mission link budgets do not share a common format and often do not state their assumptions, which makes them difficult to reproduce. A student or small research group within a university wanting to answer a specific question, for example whether a given telescope could receive from the Moon, therefore has only limited public tools to draw on.

This dissertation aims to fill this gap by presenting an open Python simulator for FSO deep-space links built from first principles, with models for atmospheric loss, turbulence and pointing loss. Every equation in it is referenced to its source, making it easier to trace and adjust if needed. The simulator is used to benchmark four published missions, to model the University of Surrey's 40 cm telescope, to perform a sensitivity analysis, to study coherent modulation with a stabilised laser, and to size a representative CubeSat-class terminal.

1.2 Objectives

The objectives agreed during the initiation of this project are as follows:

1. Build a baseline FSO link budget simulator from first principles: received power, link margin and BER for an on-off keyed link.
2. Extend it with environmental and hardware effects: atmospheric attenuation, turbulence, pointing and divergence losses, and hardware parameters from published missions.
3. Benchmark against real missions and carry out a parameter sensitivity study, to establish which parameters drive performance and by how much.
4. Investigate what is possible with a stabilised narrow-linewidth laser source by modelling coherent modulation in place of OOK.
5. Produce a component-level breakdown (hardware, mass, power and cost) of a representative FSO terminal.

1.3 Scope

The simulator is a static link budget: it gives the data rate a link supports under stated conditions, not how often those conditions occur. Cloud, season, daytime turbulence and solar conjunction geometry are outside it, as is any hardware demonstration; the stabilised-laser question of Objective 4 is answered by analysis. The benchmark set is limited to flown missions with published data, so no CubeSat-class mission is modelled end to end, and component costs, which no source publishes, are recorded as unavailable.

1.4 Achievements

The main outcomes against each of the five objectives are listed below. Not all were resolved as originally planned, but this is discussed in detail further in Chapter 5. The main achievement of the project is a working, open-source link budget simulator with an automated validation suite of 65 checks. The checks that reproduce published link budgets agree with their sources to within 0.1 dB, which confirms that the equations were implemented correctly. The environmental and hardware effects, namely atmospheric effects, pointing loss and hardware limitations, were then tested against flight data. Using only published parameters, the model predicts the change in the GOPEX uplink received signal to within 0.6 dB of the experiment's own theory and within 0.4 dB of three of the five measurements. Its turbulence model reproduces the fading loss measured on 40 cm, 1550 nm ground-station downlinks within the measured spread at 10° and 20° elevation (Section 4.2).

The mission benchmarks are accompanied by an assessment of how independent each comparison is. This assessment was not part of the original plan, but emerged from further literature review, and once the benchmarks had been run it became clear that only DSOC tests the rate model. LLCD's flown rate sits at the cap set by its modem's slot clock, so it confirms that the link closes but not the rate prediction, and TESAT is carried as a reference configuration without a measured check. Against DSOC, the simulator's decode threshold at the nominal background is 0.78 detected signal counts per symbol, about 0.7 dB below DSOC's published operating point of 0.93 counts per symbol (Section 4.3).

The planned hardware demonstration of a stabilised laser could not be completed, because delivery of the equipment to the Surrey Space Centre was delayed. Instead, a coherent receiver model was added to the simulator, which carries a finite source linewidth through to a bit error rate. The model gives the tolerable transmitter and local-oscillator linewidth at each deep-space data rate, and shows that a 100 Hz stabilised pair opens every rate above about 31 kb/s (Section 4.7). A component-level register of a representative terminal was also produced, totalling 550 g and 8.5 W and reproducing the published mass and power totals (Section 4.8).

Finally, the simulator was applied to a deep-space link into Surrey. Surrey's 40 cm telescope could receive 129 kb/s from a DSOC-class transmitter at 2.68 AU on a clear night, provided the signalling is optimised for the weaker link, using 256-PPM with slots of about 64 ns. With the signalling set DSOC demonstrated, the link closes only out to about 0.7 AU (Section 4.5, with the assumptions given in Section 4.1).

1.5 Overview of Dissertation

Chapter 2 reviews the published work that the simulator is built upon. This includes the link budget chain, the atmospheric channel, modulation and detection (specifically OOK, PPM and coherent BPSK), channel coding, and the four major missions used for validation and benchmarking on earlier simulation versions. Chapter 3 continues on from this by deriving the simulator's requirements from the objectives, describing the simulator design (the module chain and equation register) as well as the validation approach. Chapter 4 then presents the results in the order the work was done, mapped to the objectives at the start of that chapter: the forward validation, what the validation does and does not establish, inverse design, the Surrey ground station study, sensitivity, the coherent study and the sized terminal. Chapter 5 closes off by evaluating the project

against its objectives, the success or shortcomings of this project in achieving the set-out objectives and suggests further work.

2 BACKGROUND AND LITERATURE REVIEW

This simulator draws on several separate areas of the literature in order to pull together the link budget chain. Most components of the chain call for their own model and therefore literature study. For example, atmospheric model or coding model, where each needs to be introduced and explained briefly before the design in Chapter 3 can present how the equations and study is applied to this simulator. Section 2.1 introduces free-space optical communication and the notable missions that have flown it, specifically deep-space missions. Section 2.2 then gives the fundamental link budget chain that is used for this simulator. Section 2.3 continues on to review the atmospheric models available for the channel link (ground-to-space path). Section 2.4 covers modulation, detection and coding for the three receiver types that were applied by the real missions studied. Section 2.5 looks at the flown systems in more detail, in particular which of their parameters are published and which are not, because that decides what a comparison against them can prove. Finally, Section 2.6 summarises the literature review and details the gap this project addresses with the simulation and its findings.

2.1 Free-Space Optical Communication Overview

Free-space optical (FSO) communication uses an optical beam, as opposed to the traditional radio channel, as a carrier of information. The motivation behind utilising FSO has always been the fundamental physics of antenna gain against wavelength. At $1550\,\text{nm}$ the resulting advantage has been put at ten to a hundred times the data rate of an RF link, from a flight terminal of lower size, mass and power [4].

However, while this may be a reason to choose FSO over RF, optical links have disadvantages that are challenging to overcome at great distances. Table 2.1 compares these two specifically for deep-space links.

The narrow beam that gives an optical link its gain must also be held on the receiver to a small fraction of its own width: DSOC's $13.8\,\mu\text{rad}$ beam is pointed with a sub-microradian bias and $0.5\,\mu\text{rad}$ of jitter [3], which calls for active tracking and fine beam steering that a body-pointed radio antenna, aimed to about 0.1°, does not need [4]. Any residual error costs received power

through the pointing loss of Eq. (2.4). In flight this is implemented with superconducting nanowire single-photon detectors [13], and Section 2.4 describes the detection statistics that follow from it.

Table 2.1: Optical and RF deep-space links compared at $2.68\,\mathrm{AU}$. Optical column: the DSOC downlink [3]; RF column: X- and Ka-band deep-space practice [4, 9, 14].

	Optical (DSOC)	RF (X/Ka-band)
Wavelength	$1550\,\mathrm{nm}$	$0.9\,\mathrm{cm}$ to $3.6\,\mathrm{cm}$
Transmit aperture	$22\,\mathrm{cm}$	$1\,\mathrm{m}$ to $4\,\mathrm{m}$
Beam divergence	$13.8\,\mu\mathrm{rad}$	$5\,\mathrm{mrad}$ to $70\,\mathrm{mrad}$
Pointing held to	$0.5\,\mu\mathrm{rad}$ jitter	$\sim 0.1^\circ$ ($1745\,\mu\mathrm{rad}$)
Transmit gain	$112\,\mathrm{dB}$	$37\,\mathrm{dB}$ to $60\,\mathrm{dB}$
Free-space loss	$-370\,\mathrm{dB}$	$-295\,\mathrm{dB}$ to $-283\,\mathrm{dB}$
Receive gain	$140\,\mathrm{dB}$ ($5\,\mathrm{m}$ Hale)	$68\,\mathrm{dB}$ to $85\,\mathrm{dB}$ ($34\,\mathrm{m}$ to $70\,\mathrm{m}$ DSN)
Available bandwidth	Tens of THz	Hundreds of MHz
Data rate at $2.68\,\mathrm{AU}$	$8.33\,\mathrm{Mbit\,s^{-1}}$	$\mathrm{kbit\,s^{-1}}$ to low $\mathrm{Mbit\,s^{-1}}$
Capacity advantage	10 to $100\times$ at equal or lower size, weight and power	baseline
Ground network	None operational	Deep Space Network

These tradeoffs and specific equipment have been tested in real missions for two decades. In 2001, SILEX demonstrated the first inter-satellite optical link [15], and in 2006 OICETS demonstrated both inter-orbit and satellite-to-ground optical links [16].

Since then, flown demonstrations of FSO have run from inter-satellite to deep-space links: coherent inter-satellite links at $5.625\,\mathrm{Gb/s}$ (TESAT LCT, 2008) [17, 18], a lunar downlink at $622\,\mathrm{Mb/s}$ (LLCD, 2013) [8], a low-cost ISS downlink at $50\,\mathrm{Mb/s}$ (OPALS, 2014) [19], and, more recently, a photon-counting link from beyond 2 AU (DSOC, 2023-) [3]. In just over two decades, the range of feasible optical communication has moved from links between neighbouring satellites to links across Mars distances, with data returned from 2.68 AU [3]. These missions are reviewed in more detail in Section 2.5, where they serve two purposes: defining the current state of the art and providing published reference points against which the simulator of Chapter 3 is validated.

2.2 Link Budget Fundamentals

A link budget accounts for every gain and loss between transmitter and receiver to give the received power, which is then compared against the power the receiver needs to reach its target error rate. The link budget equation itself is quite simple and very well documented across the satellite communications community, but accurately calculating the individual terms is found to be the more challenging part of this problem. The received-power chain used throughout the literature

[5, 6] is

$$P_R = P_T\,\eta_T\,\eta_R\,G_T\,G_R\,L_T\,L_R\,L_A\,L_P, \tag{2.1}$$

where P_T is transmit power, η_T and η_R are the transmit and receive optics efficiencies, G_T and G_R are the antenna gains, L_T and L_R are the pointing losses at each end, L_A is atmospheric attenuation and L_P is free-space path loss [5, 6].

Link margin is the ratio of received power (P_R) to the required power (P_{req}) [5, 10]. A link is said to close when the margin is greater than 0 dB. In practice a designer would want positive margin, but 0 dB is the definition used here.

The terms that dominate a design in practice are path loss, antenna gain and pointing loss [10], and each is covered in the subsections below. The atmospheric term is more involved and has its own section, Section 2.3.

2.2.1 Free-Space Path Loss

Over a distance d, geometric spreading of the beam gives

$$L_P = \left(\frac{\lambda}{4\pi d}\right)^2, \tag{2.2}$$

which is the optical form of the Friis space loss [5, 10]. It falls as $1/d^2$, so range costs 20 dB per decade. For scale, from Eq. (2.2), a link at the Moon (3.84×10^5 km) is about 52 dB below the same link at a 1000 km crosslink, and Mars at 2.7 AU is a further 60 dB below the Moon. This single term spans more dynamic range than every other budget entry combined, and it is the main reason the design answers at different ranges in Chapter 4 differ so much.

2.2.2 Antenna and Telescope Gain

For a transmit aperture illuminated by a Gaussian beam, the on-axis far-field gain takes the form [6]

$$G_T = \frac{4\pi A}{\lambda^2}\,g_T(\alpha, \gamma), \tag{2.3}$$

where A is the aperture area, α is the ratio of aperture radius to beam radius, and γ is the central obscuration ratio [6]. The transmit beam is Gaussian, so a finite aperture always compromises: an aperture large relative to the beam (large α) passes all of the power but is under-filled and so radiates a wider far-field beam, while a small one (small α) fills the aperture but clips power at

its edge [6]. The factor g_T is the fraction of the ideal filled-aperture gain $4\pi A/\lambda^2$ that survives this trade. It peaks at 0.81 at the optimum truncation $\alpha = 1.12$ for an unobscured aperture, and a central obscuration γ, the secondary mirror of a reflecting telescope, lowers it further [6]. The receive aperture collects energy rather than forming a beam, and has the diffraction-limited gain $G_R = (\pi D_R/\lambda)^2$ for a filled aperture of diameter D_R [5]. The $1/\lambda^2$ scaling in both expressions is what produces the optical advantage described in Section 2.1. A shorter wavelength buys gain without a larger aperture: at 1550 nm the 22 cm telescope flown by DSOC reaches about 112 dB [3, 20].

When a mission publishes a beam divergence instead of an aperture, the gain can be instead written as $G_T = c/\Theta_T^2$, where Θ_T is the full divergence angle [5]. The value of the coefficient c is not fixed. It depends on the width convention used to quote the divergence, and the two conventions in common use differ by a factor of two, giving $c = 16$ or $c = 32$ ([5]; [6], Sec. III). That is a difference of 3.01 dB, which is a significant amount to lose or gain in a link budget through a definition alone. The published divergence and gain pairs of [21] are consistent with $c \approx 32$. Mission papers rarely state which convention they use, so this project records the convention alongside each mission's parameters rather than assuming one globally (Chapter 3).

2.2.3 Pointing Loss

A beam mispointed by an angle θ delivers less power on axis. For a Gaussian far field the loss follows the form

$$L_p(\theta) = \exp(-G\,\theta^2), \tag{2.4}$$

where G is the linear on-axis gain [10, 11, 5, 22]. This equation contains the main design trade-off in FSO, raising G increases sensitivity to jitter exponentially, so beyond some point a narrower beam does more harm than it does good. Under a deterministic pointing error bounded by θ_{max}, the optimum gain is given as $G_{opt} = 1/\theta_{max}^2$ ([10]). With random Rayleigh-distributed jitter, the same trade appears as an outage probability, for which Valentini et al. give closed forms [10]. Pointing error is modelled as Rayleigh distributed for zero-mean jitter and Rician for jitter about a bias ([10]).

For a uniformly illuminated aperture the far field is an Airy pattern [23] and the loss is milder: $[2J_1(\sqrt{G}\theta)/(\sqrt{G}\theta)]^2$, approximated for small angles as $\exp(-\zeta G\theta^2)$ with $\zeta = 0.188$ [10], whose exact small-angle limit is $\exp(-G\theta^2/4)$. Equation (2.4) is the convention of Valentini et al. [10]:

for a Gaussian beam of gain $G = 32/\Theta^2$ ($8/\theta_h^2$ in the half-angle θ_h) the exact small-angle loss is also $\exp(-G\theta^2/4)$, the same as the Airy small-angle limit, so Eq. (2.4) is the conservative convention rather than the physics of a Gaussian beam. It is kept for the LLCD-class and Liang-class terminals so that the Mars budget of [10] is reproduced as printed; the DSOC-class transmitter, built from its 22 cm aperture, uses the exact circular-aperture form, which reproduces DSOC's own -0.19 dB pointing allocation to within 0.02 dB (Chapter 3).

Figure 2.1 shows the net on-axis gain as a function of transmit gain for three pointing-error bounds. For each bound the delivered gain first follows the transmit gain linearly and then collapses once the exponential term takes over, so the peak is at $G_{opt} = 1/\theta_{max}^2$. Halving the pointing error gains about 6 dB at the optimum, which is a large amount in link budget terms and explains why every flown terminal carries fine-pointing hardware despite the mass cost.

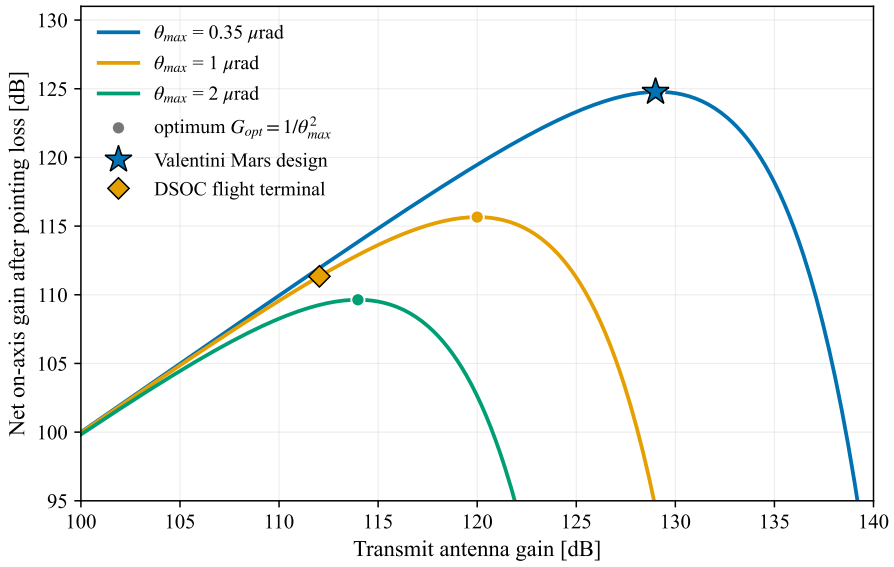


Figure 2.1: Simulator output: net on-axis gain after pointing loss against transmit gain for pointing-error bounds of 0.35, 1 and 2 μrad, Eq. (2.4). Markers: the Mars link of [10] at its optimum and the DSOC terminal [3].

2.3 Atmospheric Propagation Effects

There is no single atmospheric model that suits every case. The literature instead offers three levels of fidelity. At the simplest level, analytic Beer-Lambert extinction models are built directly from scattering physics. Where a site has been characterised in detail, an empirical model calibrated to that site can be used instead. Turbulence, which fluctuates rather than attenuating steadily, is handled separately by statistical models. Each of the above is covered in turn below.

2.3.1 Beer-Lambert Attenuation

The Beer-Lambert law describes exponential attenuation through a medium of extinction coefficient θ_A, $I = \exp(-\theta_A\, d_A)$ [5], where d_A is the propagation distance through the medium. The path length grows with zenith angle. The plane-parallel form $d_A = (h_A - h_E)\csc(\theta_E)$ ([5]) is adequate away from the horizon and air-mass formulations such as Kasten-Young [24] avoid its divergence at low elevation. The elevation dependence matters because a real pass spends much of its time at low elevation, where the air mass is largest. Even for an overhead pass from $600\,\mathrm{km}$ with a 20° mask, more than half the visible time lies below 45°, where the air mass is already 1.4 times its zenith value [25].

2.3.2 Rayleigh and Mie Scattering

Two mechanisms dominate clear-air optical extinction at the wavelengths used here with molecular absorption being negligible [26]: namely Rayleigh scattering from air molecules and Mie scattering from aerosols of size comparable to the wavelength. The Mie contribution is modelled by an elevation-dependent empirical extinction ratio fitted in wavelength, following ITU-R P.1622 [5, 27], and fog or cloud scattering with the Kim visibility model [28], in which liquid-water content and droplet concentration give a visibility, an attenuation coefficient, and a Beer-Lambert loss. No separate Rayleigh term is carried: molecular scattering falls as λ^{-4}, and at $1550\,\mathrm{nm}$ its zenith optical depth is below 0.002, under $0.01\,\mathrm{dB}$ [29]. At $1550\,\mathrm{nm}$ the clear-sky total is small, for example Liang's worked calculation gives $-0.48\,\mathrm{dB}$ at 40° elevation ([5], Table 5). Liang's tabulated values show that the geometric scattering term grows rapidly toward the horizon and eventually even reaches tens of dB in cloud.

Air mass is the quantity that carries this elevation dependence, describing how much atmosphere the beam crosses relative to a vertical path. The simplest treatment models the atmosphere as a flat slab, so the path grows as the cosecant of the elevation angle. This holds well over most of the sky, but it fails near the horizon, where it predicts an infinitely long path. The Kasten-Young formulation accounts for the curvature of the Earth and remains finite [24]. The two models agree to within a few per cent above roughly 10° and separate only below it.

2.3.3 Site-Calibrated and Empirical Models

Where a mission's ground site has been characterised directly, a site-specific model is more accurate: the DSOC mission budgets its Palomar attenuation with MODTRAN but publishes only the

resulting allocation, -0.44 dB at the design point ([3], Table III). This project therefore carries a fixed -0.5 dB clear-night allowance for the Palomar link, rounded from that allocation, and uses the Liang model only at sea-level sites: its Mie polynomial is fitted for 0–5 km but turns negative above about 1.5 km at 1550 nm, so it is clamped at zero extinction and not relied on for Palomar or Table Mountain (Chapter 3).

2.3.4 Atmospheric Turbulence and Scintillation

Scintillation is the fading of received power on millisecond timescales, caused by turbulent variations in the refractive index of the air. Its strength depends on how that turbulence is distributed along the path, which is described by the refractive-index structure profile $C_n^2(h)$. The Hufnagel-Valley model is the standard analytic form for that profile [30]. Its HV$5/7$ parameterisation takes its name from the two numbers it produces at $0.5\,\mu$m: a Fried parameter of 5 cm, which is the aperture size over which the arriving wavefront stays roughly flat [31], and an isoplanatic angle of $7\,\mu$rad, the angular field over which the distortion is essentially the same [32]. From the structure profile, weak-turbulence theory gives the Rytov variance, a single measure of turbulence strength for the whole path, and from that the scintillation index of the received irradiance [30, 32].

The forms of the profile and of the aperture-averaging factor used here, with two corrections to the secondary source, are given in Section 3.3; the Rytov variance, scintillation index and lognormal fading loss follow Eqs. 43–58 of [30].

Two refinements matter in practice: a finite receive aperture averages over the irradiance pattern and reduces the effective fading (aperture averaging; [30]) or the loss to budget depends on the required availability. Modelling the fading as lognormal converts a target outage probability into a dB allowance ([30]).

Uplink and downlink are not symmetric. A downlink arrives as an essentially plane wave and suffers scintillation only. An uplink diverges through the turbulence near the transmitter, which adds beam spread and beam wander [33]. This is modelled here as an effective divergence $\Theta_{\text{eff}}^2 = \Theta^2 + \theta_{\text{seeing}}^2$, where Θ is the divergence the beam would have in vacuum, set by the transmit optics, and θ_{seeing} is the additional angular spread imposed by turbulence above the site. The two add in quadrature because each is treated as an independent, zero-mean Gaussian angular deviation, and independent variances sum. The result is a broader beam at the far end, so received power falls as $1/\Theta_{\text{eff}}^2$ at a given range [34, 35].

Table 2.2 summarises the models reviewed in this section and where each is used in this project; the turbulence chain is compared with measurements in Section 4.2.

Table 2.2: Atmospheric models reviewed in this section and where each is used in this project.

Model	Covers	Source	Used here
Beer–Lambert, plane-parallel path	Clear-air loss	[5]	Yes
Mie aerosol polynomial (ITU-R P.1622)	Aerosol extinction	[5, 27]	Yes, sea-level sites
Kim visibility model	Fog and cloud	[28]	Yes, clear sky only
Rayleigh molecular scattering	Molecular scattering	[29]	No, negligible at 1550 nm
Site-calibrated transmission (MODTRAN)	A characterised site	[3]	Published value only
Model transmittance spectrum	Clear-sky transmittance	[36]	GOPEX only
Hufnagel–Valley HV$5/7$ profile	Turbulence strength	[30, 37]	Yes
Rytov variance, scintillation index	Fading strength	[30]	Yes, weak turbulence
Aperture averaging	Fading vs aperture	[36]	Yes
Lognormal fading at an outage probability	Loss to budget	[30]	Yes, 1% outage
Seeing added to the uplink beam	Uplink beam spread	[34]	Yes
Solar activity via NRLMSISE-00	Upper-atmosphere density	[38]	No, under 10^{-4} dB

2.3.5 Geometry

Range and elevation are not independent for a satellite in orbit; for a station seeing the satellite at elevation angle θ_E the slant range is

$$d_{GS} = R\left(\sqrt{\left(\tfrac{R+H}{R}\right)^2 - \cos^2\theta_E} \;-\; \sin\theta_E\right) \tag{2.5}$$

[5, 39]. For a station observing the ISS at 400 km, the slant range grows by a factor of 3.6 between zenith and 10° elevation, which costs 11.1 dB of free-space loss. The additional atmospheric attenuation over the same sweep is only about 1.5 dB at 1550 nm. For near-Earth links, elevation therefore acts on the budget mainly through geometry. For deep-space links the two decouple,

because range no longer depends on elevation. This point returns in Chapter 4, where it explains a documented discrepancy in one published mission budget (Section 4.3.3).

2.4 Modulation and Detection for FSO

Flown FSO systems fall into three architectures, distinguished by their detectors and therefore by their noise statistics. Each needs an error-rate model built for its own statistics, and described below. Figure 2.2 shows how each of the three carries the same bits.

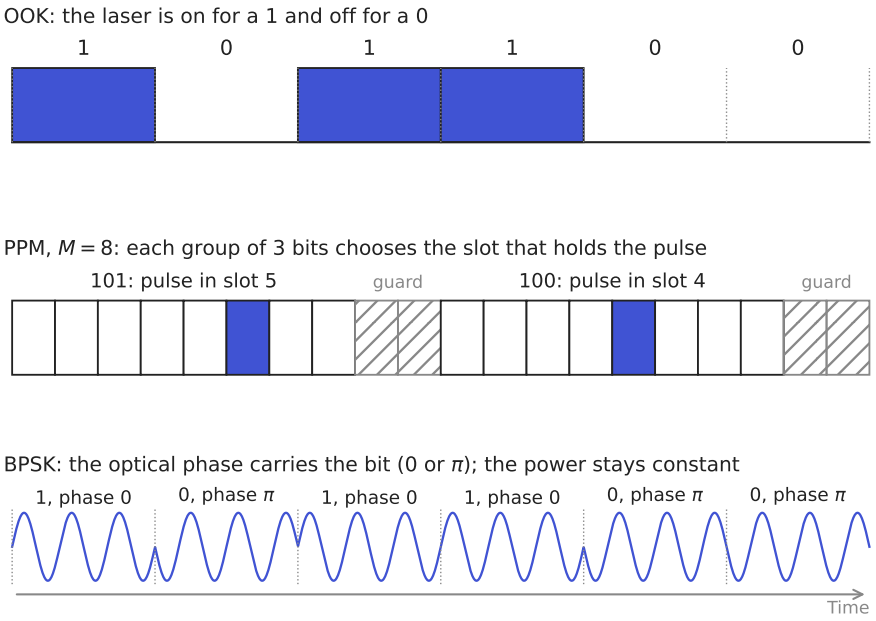


Figure 2.2: The same six bits sent as OOK, as 8-ary PPM with the $M/4$ guard slots of the CCSDS standard [1], and as BPSK. Time is not to scale between the rows, and the BPSK carrier is drawn with a few cycles per bit.

2.4.1 On-Off Keying with Direct Detection

On-off keying (OOK) switches the laser on and off once per bit and detects the result with an avalanche photodiode (APD). With many photons per bit, both signal levels are corrupted by approximately Gaussian noise: both thermal noise from the transimpedance amplifier, and shot noise amplified by the APD excess noise factor ([40], the McIntyre model). The bit error probability follows from the Q-factor,

$$p_{BE} = \tfrac{1}{2}\operatorname{erfc}\Big(Q/\sqrt{2}\Big), \qquad Q = \frac{\langle s_1\rangle - \langle s_0\rangle}{\sigma_1 + \sigma_0} \tag{2.6}$$

where $\langle s_1\rangle$ and $\langle s_0\rangle$ are the mean detector outputs for a transmitted one and zero, and σ_1 and σ_0 their standard deviations [40]. Background light enters the shot noise through the detected

background power, and the thermal and shot-noise-limited sensitivity bounds are given in the same reference. The Gaussian approximation holds from dozens of photons per bit upward [40]. OPALS is a flown example that utilised this architecture [19].

2.4.2 Pulse-Position Modulation with Photon Counting

Pulse-position modulation (PPM) is a prominent option for deep space missions, where every detected photon matters and a link cannot afford dozens of photons per bit. The laser's average power is concentrated into one pulsed slot out of the PPM-order M, encoding $\log_2 M$ bits per pulse, and detection is by a superconducting nanowire single-photon detector (SNSPD) which registers individual photon arrivals [13]. The number of photons detected in a slot follows a Poisson distribution, so its variance equals its mean and the noise is set by the signal itself. A pulsed slot has mean $K_s + K_b$ and an empty slot mean K_b, where K_s and K_b are the mean detected signal and background photons per slot [4, 10]. At DSOC's 20.6 Mb/s decode threshold the pulsed slot receives 0.93 detected signal photons on average, so it is still empty $e^{-0.93} \approx 39\%$ of the time; individual symbols are often lost and the code has to recover them (Section 2.4.5). Hemmati et al. give the exact maximum-likelihood symbol error probability [4]; with no background this reduces to $P_s = (M-1)e^{-K_s}/M$, and the simulator implements the full form with background ([4], Eqs. A5–A7). Symbol errors map to bit errors as $P_b \approx \frac{M}{2(M-1)} P_s$ [41].

Photon efficiency grows with M while the slot rate grows as $M/\log_2 M$, so bandwidth is being traded for photon efficiency. How far this can be pushed is limited by detector timing (Section 2.4.4). LLCD and DSOC demonstrate examples of PPM usage in flown missions. [8, 3].

Evaluating the exact decision rule of [4] at a night-time background of 0.01 counts per slot, a 16-ary symbol reaches a 10^{-4} bit error rate on about 9 detected photons per symbol, i.e. a little over two photons per bit. Each doubling of the order gives further sensitivity at the cost of a longer, slower symbol, which is why photon-starved links run at high orders and low duty cycles.

2.4.3 Coherent BPSK with Homodyne Detection

Coherent detection abandons intensity modulation and uses the optical phase to carry information. Binary phase-shift keying (BPSK) encodes bits in the phase of the optical carrier, recovered by mixing the received field with a local-oscillator laser. In the shot-noise limit, homodyne BPSK achieves

$$\mathrm{BER} = \tfrac{1}{2}\,\mathrm{erfc}\big(\sqrt{2\bar{n}_s}\big)\,, \tag{2.7}$$

where $\bar{n}_s$ is the mean detected photons per bit; this gives 4.55 photons per bit at a BER of 10^{-5} [42]. Heterodyne and intradyne reception pay a 3 dB image-band penalty ([42]). This is the best photon efficiency of any uncoded scheme at useful bandwidth efficiency, and it is the scheme flown by the TESAT terminal [17, 18]. However, the transmitter and local-oscillator linewidths add, and the residual phase-error variance of the receiver's tracking scheme degrades the BER and sets an error floor; because the phase drifts furthest within a long symbol, affecting low data rates more significantly [43]. For a transmitter and local oscillator of combined linewidth Δf, the phase drifts over a bit period T by a Gaussian amount of variance $\sigma_\phi^2 = 2\pi\,\Delta f\,T$ ([43], Eq. 60). The heterodyne DPSK error rate with that residual phase error is Eq. (36) of [43], a triple integral over the two Rician decision envelopes and the Gaussian phase error, which the simulator evaluates numerically; it reduces to $\frac{1}{2}e^{-\gamma}$ at $\sigma_\phi = 0$, with γ the detected photons per bit, and to the floor $\operatorname{erfc}\big(\pi/(2\sqrt{2}\,\sigma_\phi)\big)$ at high signal-to-noise ratio. A laser line narrow and stable enough to make that variance negligible is what separates the stabilised source of Objective 4 from a standard communications laser. Equation (2.7) sets an ideal bound by assuming perfect phase tracking and no implementation loss, so a real receiver falls short of it.

Figure 2.3 compares the three schemes at the quantum limit, with no background. At a 10^{-4} bit error rate, 256-PPM needs 1.1 detected photons per bit and 16-PPM 2.1, against 3.5 for homodyne BPSK, 4.3 for OOK and 6.9 for heterodyne BPSK. Heterodyne detection needs exactly twice the photons of homodyne. Recovering that factor requires a phase-locked receiver, and so a stabilised laser (Section 4.7).

2.4.4 Achievable Rate: Optimising the Signalling Format

Given a received photon flux, the supportable data rate cannot be assumed; it has to be found by optimisation. This was not obvious at the start of the project and it is one of the ways a deep-space budget differs from a near-Earth one. The inverse question is direct: for a chosen signalling configuration, the flux needed to decode at a given rate follows from the capacity expression, and the design results of Chapter 4 use it that way. The forward question is not, because the best rate depends on which PPM order, slot width and code rate are chosen, so it has to be found by searching over that set. The method used by JPL's deep-space link tool [12] optimises PPM order, code rate and slot width against the capacity of the Poisson PPM channel, subject to the receiver's minimum slot width. However there are two consequences of this for this project.

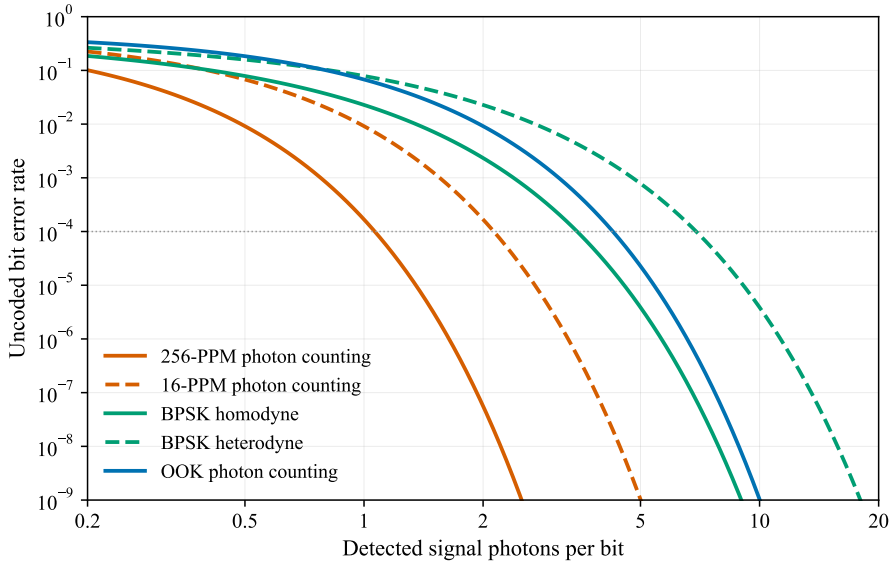


Figure 2.3: Simulator output: uncoded bit error rate against detected photons per bit for photon-counting M-ary PPM and OOK [4] and shot-noise-limited homodyne and heterodyne BPSK [42], all without background.

First, the slot width sets an upper limit on rate. For any link with power to spare, the rate is capped by how fast the detector can resolve slots, and additional received power does not raise it. Xie et al. conclude that detector speed rather than aperture will limit future deep-space rates [12]. Second, closing the optimisation against capacity separates the channel physics from the performance of any particular code (Section 2.4.5).

The CCSDS optical coding standard [1] defines the format that was flown: SCPPM codewords of 15 120 bits at rates 1/3, 1/2 and 2/3, PPM orders 4 to 256, and a guard time of $M/4$ slots between symbols. The guard time is a small detail, but it is what reconciles the published data rates with the slot arithmetic ([10], Tables I-II). The frame information rate is

$$R_b = \frac{15120\,R_c - 34}{(15120/\log_2 M)\;M\,T_s\,(1+\alpha_{gt})}, \tag{2.8}$$

with R_c the code rate, T_s the slot time and $\alpha_{gt} = 1/4$ the guard fraction ([10], Eq. 20; [1]); the printed form of that equation omits the "1+", and only the form above reproduces the paper's own tabulated rates.

2.4.5 Forward Error Correction and Coding Gain

The soft-decision capacity of the M-ary PPM Poisson channel is computed from the slot likelihood ratios [44, 45], acting as the ceiling against which real codes are measured. The SCPPM code of the CCSDS standard operates approximately $1\,\mathrm{dB}$ from that capacity at a BER of 10^{-5}, with LDPC alternatives at approximately $1.4\,\mathrm{dB}$ ([4], Sec. III; the code itself is defined in [1] and its gap to capacity quantified in [44]). Coded performance can therefore be modelled in one of two ways: as capacity evaluated with the received flux reduced by this gap, or as a coding gain applied to received power in a BER-target calculation. The two must never be combined, because capacity already assumes ideal coding and applying a coding gain on top double-counts it. The option chosen for this project is discussed further in Chapter 3.

The value of coding is clearest on a real link. On a photon-limited link, coding has the ability to multiply the achievable rate nearly 7 times over at 1AU on a DSOC-class downlink; Chapter 4 quantifies this in full (Figure 4.3).

2.5 Review of Real Flown FSO Systems

Table 2.3 summarises the flown systems used as benchmarks. They span three orders of magnitude in range and four in data rate, and between them cover all three detection architectures of Section 2.4, which is why they were chosen. For each mission the relevant question is not just the headline performance but which parameters were published and which were not. This decides what a comparison against that mission can actually prove, and it becomes a main theme of Chapters 3 and 4.

Table 2.3: A review of flown FSO systems used as benchmarks in this project.

Mission	Year	Link	Range	Rate	Modulation / detector
TESAT LCT [17, 18]	2008	LEO-LEO	$\leq 5100\,\mathrm{km}$	5.625 Gb/s	BPSK homodyne
LLCD [8, 46]	2013	Moon-ground	$4 \times 10^5\,\mathrm{km}$	622 Mb/s	16-PPM / SNSPD
OPALS [19, 47]	2014	ISS-ground	400–900 km	50 Mb/s	OOK / APD
DSOC [3]	2023-	Deep space-ground	0.37–2.68 AU	≤ 267 Mb/s	PPM / SNSPD array

Placed together as published data rate against link distance (Figure 2.4), two decades of flight teach two things. First, each detection architecture is only flown over a particular range of distances: OOK with an avalanche photodiode at LEO ranges, coherent BPSK on crosslinks, and photon-counting PPM everywhere beyond the Moon. Second, once a link is photon-limited, range sets the

rate: DSOC's flown rates fall as $d^{-1.8}$ between 0.56 and 2.68 AU, close to the $1/d^2$ of free-space path loss (Section 2.2), and the small departure from $1/d^2$ is the discrete signalling set, which lets the rate change only in steps.

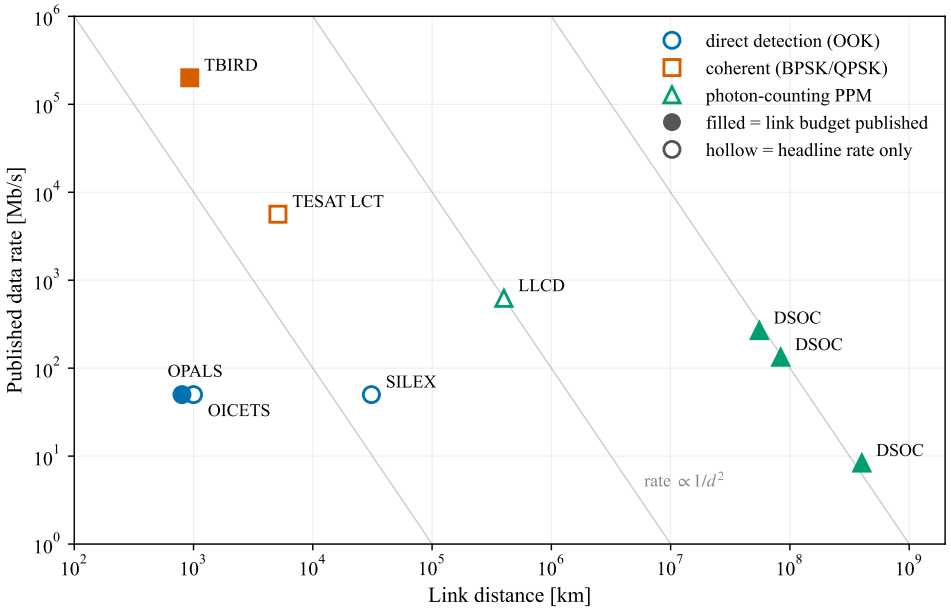


Figure 2.4: Published data rate against link distance for the flown FSO demonstrations reviewed in this section [15, 48, 17, 47, 8, 49, 3]. Marker shape gives the detection architecture; filled markers mark missions that published a link budget. Grey guide lines show a constant rate times distance squared, the $1/d^2$ fall of free-space path loss.

2.5.1 LLCD (2013)

The Lunar Laser Communication Demonstration (LLCD) flew on LADEE in 2013 and was the first optical link closed from lunar distance. It achieved 622 Mb/s downlink over approximately 4×10^5 km using 16-ary PPM with SNSPD receivers on the ground [8]. The flight terminal was small, a 10 cm aperture and 0.5 W transmit power [50]. Whilst LLCD published its architecture and headline rates, it did not publish a full link budget. Apart from the ground receiver's published sensitivity [46], its receiver-chain parameters are therefore treated as derived quantities in this project rather than published values, and Chapter 4 explains what this means for the comparison.

2.5.2 DSOC / Psyche (2023-)

DSOC is flying on the Psyche spacecraft since its launch in 2023, and is currently the most advanced deep-space optical link. The flight terminal has a 22 cm aperture and 4 W average transmit power, and the signal is received at the 5 m Hale telescope at Palomar using SNSPD arrays [3]. The system reached 267 Mb/s at 0.37 AU and still delivered 8.3 Mb/s at 2.68 AU [3].

For this project DSOC is by far the most useful benchmark. The mission team has published a more complete account than is usual, including link budget tables, received-power predictions and measured data rates across the mission range [3]. It is the only mission in the set for which an independent end-to-end comparison is possible, and much of Chapter 4 depends on it.

2.5.3 OPALS (2014)

The Optical Payload for Lasercomm Science (OPALS) downlinked video from the ISS at 50 Mb/s using OOK and an APD ground receiver. It was deliberately low-cost and built largely from commercial parts [19]. OPALS published a link budget with a stated margin [47], which makes it one of the few missions offering a number a simulator can be checked against directly, which is described further in Chapter 4.

2.5.4 TESAT LCT (2008)

The TESAT laser communication terminals on TerraSAR-X and NFIRE demonstrated coherent BPSK homodyne crosslinks at 5.625 Gb/s between LEO satellites at a BER of 10^{-9} [17], with the same terminal line measured on a satellite-to-ground link [18, 51].

The same terminal line now serves operationally in the European Data Relay System. As the only flown coherent system in the benchmark set, it anchors the stabilised-laser study of Objective 4. Its published record consists of headline performance rather than budget detail, so its receiver sensitivities are treated as derived quantities rather than validation targets.

2.5.5 Small-Satellite Terminals

At the small end of the mass scale, the NODE design study specified a CubeSat-scale terminal including transmit aperture, fine steering, and mass and power budgets for a sub-kilogram package [52]. NODE is the basis for the representative terminal sized in this project (Objective 5). In flight, TBIRD demonstrated 200 Gb/s burst downlinks from a 6U CubeSat in 2022 [53], confirming that optical downlinks fit CubeSat mass and power budgets. This project does not model a CubeSat-class mission end-to-end since the benchmark set is deep-space and flagship-LEO, where published validation data exists. This decision is discussed further in Chapter 3.

The benchmark sources differ most in how much of their flown data is published. Table 2.4 lists, for each, what was published, what was not, and what that leaves a comparison able to test. The unpublished values are carried as ASSUMED in this project (Chapter 3).

Table 2.4: What each benchmark source publishes and leaves out, and what a comparison against it can therefore test.

Source	Published	Not published	A comparison can test
Liang ISL and ground links [5]	Complete worked budgets: power, divergence, apertures, efficiencies, pointing, atmosphere and receiver sensitivity	Nothing the arithmetic needs; the budgets are design calculations, not measurements	The implementation (reproduction only)
GOPEX uplink [34]	Range, beam divergence, seeing, zenith angle and received-pulse statistics for each night	Camera aperture and quantum efficiency; atmospheric transmission on each night	The night-to-night change in received signal, relative to day 1
LLCD [8]	Transmit power and aperture, receive array, detection efficiency, PPM order, slot clock and data rate	Optics efficiencies, pointing error, code rate, background and code gap	Closure only: the rate is set by the slot clock
OPALS [19, 47]	Data rate and a link budget with a stated margin	Receiver-side camera constants	No measured check; a reference configuration
TESAT LCT [17]	Data rate, modulation and bit error rate	Link budget and receiver sensitivity	No measured check; a reference configuration
DSOC [3]	Link budget with design, favourable and adverse values; measured rates at three ranges; background range; code gap	The modem's signalling set beyond the flown configurations; the site atmosphere model	The rate model, at the two power-limited flight points

2.6 Summary

Several points from this review shaped the design in Chapter 3. The link budget chain is standard and each of its terms is well documented, but published treatments differ in their conventions, for the divergence width and for the gain definition, by up to 3 dB, so a simulator has to state which conventions it uses. Atmospheric models exist at several fidelities and none is best in every case; the right choice depends on the site and on the question, and may differ between links in the same study (Table 2.2). The three detection architectures have different noise statistics, so the error model has to follow the architecture. The flown missions provide performance anchors across the design space, but their publications are incomplete: headline rates are public while the receiver chains and the calculations behind the budgets are not (Table 2.4), so validation against them has to separate what is being tested from what is being assumed, a distinction the literature does not always make. Finally, the tools behind published deep-space link budgets, most notably

JPL's SOLT [12], are described but not available, and no open tool was found that reconstructs their function with every equation and parameter traceable to a published source. Two recent MSc theses, [30] and [45], build deep-space link models and are drawn on here as sources. Neither publishes a validated, reusable tool with its equations and parameters traced to primary sources, which is part of the motivation for the equation register of this project.

It is this last gap that the project addresses: an open deep-space link budget simulator, benchmarked against the values published for flown missions. Where those publications leave parameters unstated, assumptions are made, but each is stated and its choice explained.

3 DESIGN AND METHODOLOGY

Chapter 2 ended on the observation that although the physics and the flown designs of deep-space optical links are well documented, the tools that combine them into a link budget are not. This chapter presents the simulator built in response to the lack of this tool, and the method by which it is built. Section 3.1 states the requirements derived from the objectives, Section 3.2 describes the design and implementation, Section 3.4 covers the link budget and error-rate implementation choices, Sections 3.5 and 3.6 cover the phase-noise analysis (Objective 4) and inverse design (the design questions of Objective 3), Section 3.7 presents the validation methodology, Section 3.8 describes the Surrey ground station question and Section 3.9 states how the results are regenerated.

3.1 Requirements

This project's objective is to implement a link budget simulator for free-space optical satellite communications, extendable from a simple on-off keyed link to the photon-counting and coherent regimes of lunar and deep-space communications. As established in Section 2.6, the tools that generate the published link budgets of the field are not available to the research community since mission papers present selected figures from a proprietary model, and the established deep-space tool, JPL's SOLT, cannot be obtained [12]. The simulator described in this thesis is therefore designed to be fully auditable from the published literature, with every equation traceable to a published source.

The five objectives are stated in Section 1.2. The requirements on the simulator are built from them.

Every equation implemented in the simulator is recorded with its source (paper and equation number), symbol definitions and units before it appears in code. All internal calculations use linear SI units, and decibel quantities appear only at the input and output boundary; this removes a common class of transcription error in link budget work, where dB and linear forms of the same quantity are mixed. Each model must pass an automated check (the validation suite) before its outputs are used in results, either against a published worked example where one exists, or against flight measurements where mission data permit. In addition to the forward question, given a link, what data rate results, the simulator must answer the design question: given a distance and a required data

rate, what transmit power or transmit aperture closes the link. Finally, the simulator must carry parameter sets for flown missions, with every value cited to its source and unpublished values explicitly marked as assumed, to allow for reproducibility to those with access to the data.

3.2 Design Approach

The simulator is written in Python as a set of modules, each representing one stage in the received-power budget of Eq. (2.1). These are combined into a forward model that predicts received power. Received power is converted to photon flux and passed to one of two receiver models: Poisson statistics for PPM/SNSPD [4], or the shot-noise-limited coherent model [42]. The receiver model output is converted to a supported data rate by the modulation and coding model. Figure 3.1 shows the module chain. OOK links are closed against a published receiver sensitivity (Liang's -35.5 dBm at 10 Gb/s [5]) rather than a detector noise model, since no result in this dissertation depends on the APD noise chain.

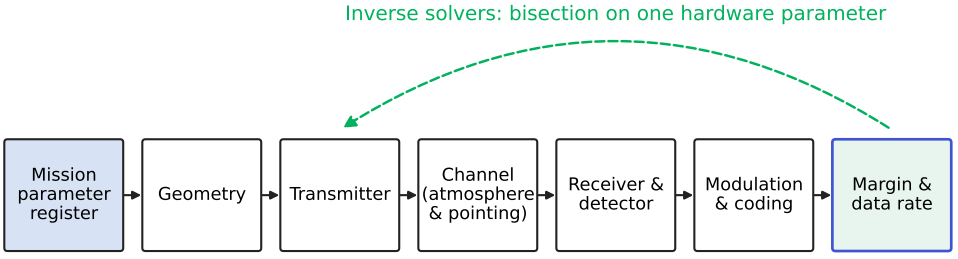


Figure 3.1: Simulator module chain: the forward path of Eq. (2.1) in linear SI units, and the inverse solvers that bisect on it.

As a result of every module returning a linear value, the same forward model can be used in an optimisation to solve for a design goal. The value of a single design parameter that meets a target data rate is found by bisection. This is used throughout the inverse design work in Section 3.6, with some care needed around the non-monotonic region of transmit aperture. Second, because mission parameters are data rather than hard-coded, the same model covers everything from LEO inter-satellite links to deep-space downlinks beyond 2 AU. This was done for convenience but also proved a useful check on the implementation.

Published mission values are usually incomplete, particularly receiver efficiency, which is almost never given in full. To handle the unknown losses mentioned in published papers, it was decided that they would be cancelled in a comparison (e.g. a ratio of data rates between two operating

points of the same missions, used for the GOPEX mission for a night-by-night comparison). This approach is followed and discussed in Chapter 4.

3.3 Design Choices

Each point below selects a particular option from those discussed in Chapter 2, and explains the reasoning behind the choice.

3.3.1 Atmospheric Model Selection

Atmospheric loss is attached to the ground site. Every station uses the Liang clear-air model, the Mie polynomial plus the Kim fog term [5, 28], evaluated at the site altitude and the link elevation; the model reproduces Liang's Table 5 to within $0.005\,\mathrm{dB}$. The polynomial is fitted for station altitudes of 0–5 km but turns negative above about $1.5\,\mathrm{km}$ at $1550\,\mathrm{nm}$, so it is clamped at zero extinction, and for the Palomar link the DSOC budget's own $-0.44\,\mathrm{dB}$ allocation ([3], Table III), rounded to $-0.5\,\mathrm{dB}$, is carried instead. Running MODTRAN was considered and rejected as an undocumented dependency. In the simulator the atmosphere is opt-in: a link budget carries no atmospheric term unless a station and an elevation are given, so vacuum crosslinks cannot pick one up by accident, and every ground-station scenario in Chapter 4 supplies both.

Turbulence follows the chain of [30]: Hufnagel-Valley profile, Rytov variance, scintillation index, aperture averaging and a lognormal fading allowance at a stated outage probability. That thesis is used as a collected statement of the chain; each equation in it is traced to its primary source ([32], [36], [37]), and where the two differ the primary source is followed. The two equations concerned are the profile,

$$C_n^2(h) = 0.00594\left(\frac{w}{27}\right)^2\left(10^{-5}h\right)^{10}e^{-h/1000} + 2.7\times10^{-16}\,e^{-h/1500} + A\,e^{-h/100}, \qquad (3.1)$$

with $w = 21\,\mathrm{m/s}$ and $A = 1.7\times10^{-14}\,\mathrm{m}^{-2/3}$ for HV$5/7$ [37, 32], and the downlink aperture-averaging factor for a receiver of diameter D_R at zenith angle ζ,

$$\Lambda = \left[1 + 1.11\left(\frac{D_R^2\cos\zeta}{h_s\,\lambda}\right)^{7/6}\right]^{-1}, \qquad (3.2)$$

with h_s the turbulence scale height ([36], after [54]). Checking the implementation against the primary sources showed two places where the secondary statement of these equations departs from them, and both are corrected in the simulator. First, the height h in Eq. (3.1) is height above

the local ground [37]. The path integrals run in height above sea level, so the profile is evaluated at the height above the station; evaluated at sea-level height instead, the ground layer disappears at any elevated site, and the modelled Fried parameter at a mountain site comes out three to four times the measured value. Second, Eq. (3.2) has $\cos\zeta$ in the numerator, whereas [30] places it in the denominator. The printed form makes averaging strengthen toward the horizon, which is the opposite of the trend measured on the KIODO downlinks [48]. Both errors came to light when the model was compared with measurements (Section 4.2): the averaging error through the fading loss at low elevation, and the height error through the Fried parameter at mountain sites. Two further differences between the sources could not be settled from the literature, and are discussed in Section 5.3.

Three other treatments of the atmosphere were tried and set aside. A first-principles clear-air model, Rayleigh scattering from tabulated optical depth [29] with an aerosol term after Kim [28], was built first and replaced by Liang's empirical fit once the validation suite had been written against Liang's tables, whose fit was more accurate (0.005 dB) in reproduction; the Rayleigh term is in any case below $0.01\,\mathrm{dB}$ at $1550\,\mathrm{nm}$. A site-calibrated Palomar model was likewise built and withdrawn when it was found to double-count turbulence that the fading allowance already carried, in favour of DSOC's published allocation. The Kasten-Young air mass [24] was evaluated against the plane-parallel path: the two differ by less than $0.01\,\mathrm{dB}$ above 20° and by $0.05\,\mathrm{dB}$ at 10°, beneath every operating point in Chapter 4, so the simpler form was kept. Finally, the effect of solar activity on the clear-air loss was examined with the NRLMSISE-00 model [38] at Palomar's coordinates. Between solar minimum and maximum (F10.7 indices of 70 and 200, Ap of 5 and 80) the column density between 50 and 500 km changes by $0.02\,\%$, and that band holds about a thousandth of the mass of the troposphere, so the effect on the loss is below $10^{-4}\,\mathrm{dB}$ and is not modelled: a checked negative result rather than an unexamined assumption.

3.3.2 Modulation and Detection Schemes

The three schemes modelled in this project have been selected because they each represent a distinct receiver architecture that has been demonstrated in flight (Section 2.5): OOK with an avalanche photodiode (OPALS), PPM with photon-counting SNSPDs (LLCD, DSOC), and coherent BPSK with homodyne detection (TESAT LCT). Each has its own noise model and BER calculation, so all three are modelled and the receiver architecture selects which applies. Published measurements of detector behaviour were reviewed for inclusion: SNSPD timing jitter and dead

time [13, 55], an uncoded 16-PPM error curve measured on flight-like hardware [56], and receiver implementation loss [57]. None is carried separately, because every coded result in Chapter 4 is derated by DSOC's measured 1.8 dB gap between capacity and flight decoding, which already contains these effects for the one receiver that is tested against flight data; adding them again would count them twice. An external-cavity laser class at 10 kHz linewidth was dropped from the coherent study for the plainer reason that no source for the figure could be found, leaving the published DFB range [42] and the stabilised source specified for this project.

3.3.3 Mission Parameter Sets

Each parameter in the mission parameter sets includes a reference to its source, and parameters with no stated source are annotated as assumed. The four flown missions of Table 2.3, plus the Liang inter-satellite link scenario used for arithmetic reproduction [5], are used as a parameter set. This project does not model a CubeSat-class mission (see Section 2.5): the four missions are used as a set as they represent the most published data available, including ground site information, while CubeSat-class missions publish the least. Two further choices were made here. TBIRD [53] was considered as a fifth flown system and left out, because no configuration the simulator models reaches its operating point, so a comparison would have tested nothing. OPALS and TESAT were at first treated as measured benchmarks and are carried instead as reference configurations, since neither publishes the receiver constants that a rate reproduction would need (Section 4.3).

The baseline values for the two mission configurations behind the results of Chapter 4, the DSOC downlink as flown and the LLCD-class lunar downlink, are carried in the parameter register with a citation against each value; the DSOC values appear in the worked budget of Table 3.1, and the Surrey station is specified in Table B.1 (Appendix B). Liang's configurations are used exactly as published ([5], Tables 2 and 5).

Where a result depends on an assumed value, the value is carried as a range rather than a point value. Each range spans the hardware that could plausibly fill that role: detection efficiency, for example, runs from 0.2, typical of an InGaAs avalanche photodiode, to 0.85 for a good superconducting nanowire detector [13], a span that includes the 0.5 of LLCD's detectors [8] and the 0.72 of DSOC's [3]. The ranges are bounds rather than probability distributions, because no distribution is published for these parameters. Each parameter moves the result in one direction only, so evaluating the model once with every range at its pessimistic end and once at its optimistic end brackets the answer [58].

3.4 Link Budget and Error-Rate Implementation

3.4.1 Link Budget Implementation

The received-power budget of Eq. (2.1) is calculated term by term in linear units, with the dB budget only being assembled at the end. The link geometry provides the distance and for near-Earth links the slant range is calculated from the altitude and the link elevation angle using Eq. (2.5), while for deep-space links the range is provided as input. Transmit gain is calculated from the Klein and Degnan aperture model of Eq. (2.3) [6] when the aperture of the transmitting terminal is published, or from the divergence form $G_T = c/\Theta_T^2$ [5] when only the beam divergence is published. The coefficient c is 16 or 32 depending on the width convention, which is recorded for each mission (Section 2.2). Table 3.1 shows the values for one mission, at one operating point.

Table 3.1: Simulator output: worked gain/loss budget for the DSOC downlink at the 2.68 AU post-conjunction operating point (4 W transmit). Assumed parameters as marked.

Term	Basis	Value
Transmit power	4 W	+36.0 dBm
Transmit antenna gain	22 cm flight telescope	+112.0 dB
Transmit optics efficiency	0.673, [3] Table III	−1.7 dB
Transmit pointing loss	1 μrad bound, [3] Sec. III.E	−0.2 dB
Free-space path loss	2.68 AU at 1550 nm	−370.2 dB
Receive antenna gain	5 m Hale telescope	+140.1 dB
Receive chain efficiency	0.30, [3] Table III (0.25–0.38)	−5.2 dB
Other losses		−0.5 dB
Received power		−89.7 dBm

3.4.2 BER Estimation per Modulation

The error model for each receiver architecture is selected based on the noise characteristics of the architecture: the exact maximum-likelihood solution for Poisson noise in PPM/SNSPD [4], and the shot-noise-limited bound of Eq. (2.7) for coherent BPSK [42]. The OOK links of Chapter 4 are closed against a published sensitivity, so the Q-factor model of Eq. (2.6) is not exercised. Of the two ways of applying coding described in Section 2.4.5, this simulator uses the capacity-gap form. Every coded result in Chapter 4 is a data rate, and this form gives that rate directly, as the capacity at the received signal level less a fixed gap. The coding-gain form instead gives the error rate at a data rate that has already been chosen. Finding a rate with it would need a search over rates and a separate coding gain for each code rate and PPM order considered. The gap used is also a measured one: DSOC's 1.8 dB [3] comes from a flown link, so it includes the synchronisation losses of a real receiver as well as those of the code. The PPM capacity is evaluated by Monte

Carlo simulation over the Poisson slot statistics, with 20 000 samples per evaluation and a fixed random seed; across four seeds the DSOC decode threshold of Section 4.3 moves by about 0.1 dB, which sets the precision of the coded results. The forward rate search follows Algorithm 1 of [12]: for each code rate in $\{1/3, 1/2, 2/3\}$, each PPM order in the signalling set and each slot width from the longest downward, (1) the signal and background counts per slot, K_s and K_b, are formed from the detected flux, the background and the dark counts; (2) the capacity of the Poisson PPM channel [44, 45] is evaluated with K_s reduced by the 1.8 dB gap; (3) the configuration closes if the code's information rate in bits per slot does not exceed that capacity; and (4) the highest frame rate of Eq. (2.8) among the configurations that close is returned, an order being abandoned at the first slot width that fails, since shorter slots only make closing harder.

3.4.3 Limits of the Idealised Formulas

The idealised formulae have no implementation floor, so at high margins they return error rates that have no basis in reality; such values are therefore reported as theoretical bounds only. The PPM symbol error expression also has a numerical validity limit at high photon numbers. However, real operating points for the mission benchmarks are at low photon numbers where the expression is valid, and so no calculation in this thesis is affected by this limitation. The limit is noted with the formula's description because checking validity limits is a part of using these formulae correctly. Where a published measurement exists it is used in place of an ideal model: DSOC's receive-chain losses ([3], Table III) replace an assumed efficiency, and its measured 1.8 dB gap between capacity and flight decoding replaces an ideal code. Detector dead time is not modelled; the photon-counting receiver is taken as an array with no blocking at the count rates reached, as DSOC's is, which matters only for the near-Earth rows of the Surrey table.

3.5 Stabilised-Laser Coherent Study

Objective 4 of this dissertation asks what performance can be expected from an optical communication link when the optical source is not a communications laser, but a stabilised narrow-linewidth laser of the type used in precision photonic instruments. The choice to include an experiment with a stabilised laser source was made at the project's initiation. However, testing with this equipment was eventually deferred due to a delay in delivery of the parts required to conduct such a test, and this dissertation instead addresses it through analysis and simulation. The method used is the phase-noise chain of Kikuchi *et al.* [43]: a linewidth Δf produces a phase wander of variance $2\pi\,\Delta f\,T$ over a period T; each receiver architecture has a conversion from linewidth to residual

phase variance (their Sec. III); the DPSK error rate with that residual phase error reduces to $\frac{1}{2}e^{-\gamma}$ without phase noise and to an error floor set by the phase error alone at high signal-to-noise ratio [43]. Sweeping linewidth against data rate identifies the operating threshold between normal and stabilised lasers and is the basis of the comparison in Chapter 4. The comparison is conservative, as the coherent model is the shot-noise-limited receiver, and any advantage of a stabilised source is therefore an advantage over the best possible alternative.

3.6 Inverse Design and Terminal Sizing

This project's inverse solvers work by running the forward model repeatedly rather than by rearranging it since the functions have no closed-form inverse. For a coded PPM link, the detected photon flux that just decodes at a target rate follows from the capacity expression of Section 2.4.5. One hardware parameter is then chosen as the unknown, all the others are held fixed, and a bisection search finds the value of that unknown at which the forward model delivers that flux. Two parameters are solved this way in Chapter 4: transmit aperture at a fixed laser power, and laser power at a fixed aperture.

The search starts from an interval known to contain the answer and narrows it until the delivered flux lands on the target. This is only valid if the flux moves in a single direction as the parameter varies, so that exactly one crossing exists. That holds for power. It does not hold for transmit aperture, where the pointing-loss trade of Section 2.2 means the delivered flux first rises and then falls, giving two possible solutions; the search is therefore restricted to the physically plausible branch, below the pointing-limited optimum.

One further detail of the solver becomes important in Chapter 4. A mission that uses coding must be sized against its coded error rate rather than its uncoded one. If the uncoded budget is applied to a coded mission, the model is effectively asked to buy in hardware the gain the code would have supplied which is paid for by the receiver aperture diameter, and Chapter 4 quantifies this penalty for the DSOC baseline.

For Objective 5, the component list for the representative terminal is taken from the NODE CubeSat design [52]. Each component carries its mass and its power, together with a note recording whether the value was measured, estimated, or allocated.

3.7 Validation Methodology

Every model is subjected to automated checks before being used to generate results. The checks fall into two categories which differ in what they can prove. Firstly, reproduction checks recompute a worked example from a published paper, to within a tolerance set by the precision of the published value, and so confirms that the equations have been implemented correctly. The sources are listed in Table 4.1, what each publishes is set out in Table 2.4, and Section 4.1 explains how each tolerance is chosen. Secondly, a measured check compares the model with measurements that were not used in its construction, and so tests whether the simulation describes real links rather than whether its equations were entered correctly.

Several mistakes and regressions were caught this way during development using both categories, specifically reproduction when adding new models to the simulation. The tolerances were first written as percentages and were changed back to absolute differences in each quantity's own unit when the suite grew to include checks whose expected value is zero, for which a percentage is undefined; Section 4.1 gives the tolerance used for each kind of check and the reason for it.

Where a published source gives a result only as a graph, values are read off the figure and entered as numerical reproduction checks, with a tolerance set by an estimated reading accuracy. The PPM channel capacity behind every coded deep-space rate is checked this way against Fig. 5 of [4]: at eight points across PPM orders from 2 to 256, the model reaches the same capacity at the same received power to within 0.2 dB. The DPSK error rate under phase noise is checked in the same way against Fig. 3 of [43] (Table 4.4). The comparisons with flight measurements are presented in Section 4.2.

The errors these checks caught while the codebase was being built, and what a passing check does and does not prove, are set out with the validation results in Section 4.2.3.

3.7.1 Distinguishing Evidence from Arithmetic

A comparison that reproduces a published value only counts as evidence of verification if that value was not itself used to build the model, known as circular verification. A link is power-limited when the number of received photons sets its achievable data rate, and bandwidth-limited when there is power to spare and the fastest timing its hardware supports, its slot clock, caps the rate. This matters here because several of the benchmark links are bandwidth-limited. Their symbol or bit rate had to be derived from their own published data rate, so a comparison of the

model's rate against the published value only reproduces that arithmetic, meaning it cannot fail. What such links do test is closure, i.e. whether the power budget supports the published rate at all, and this is still worth noting.

For this reason the validation results in Chapter 4 are split into power-limited cases, which test the rate model, and bandwidth-limited cases, which test closure only. This distinction was not apparent at the start of the project. It became clear once the benchmarks were run and one of them agreed suspiciously well (Section 4.3). In short, a mission whose flown rate sits at its modem's slot-clock cap is reproduced by any model that closes the link, however wrong its power budget, so the reproduction proves nothing about the budget. Section 4.3 sets out the argument mission by mission, including the LLCD case, in which, during development, the published rate survived an artificial 13 dB penalty inserted into the budget. With the budget as it now stands, whose LLCD margin is 10.1 dB, that penalty would leave the link short of closing, so the observation records how the circularity was found rather than a check that can be re-run.

3.8 Surrey Ground Station Study

The design question of Section 3.6 is applied to a specific case: what the University of Surrey's existing telescope could receive as an optical ground station across LEO, lunar and deep-space ranges, with the deep-space case developed in detail, and what transmitter each would require. The question was chosen partly because it is a natural one for the University and partly because it exercises the inverse solvers on a link outside the benchmark set. The station is the University's Orion Optics ODK16, a 40 cm aperture with a 152 mm secondary mirror, based in Guildford [59]. Optics throughput, detector quantum efficiency, dark counts and site seeing have not been measured and are all marked assumed. The link budget is solved in the inverse direction for the power and aperture requirements of a candidate transmitter at each range. The limitations of this, which are considerable, are discussed in Chapter 4.

The station parameters are listed in Table B.1 (Appendix B). Only the aperture and central obscuration are taken from the manufacturer's specification. It is not known if the station has an optical communications receiver, so every detector and back-end value is assumed, each with an interval spanning plausible hardware; results in Chapter 4 are quoted at the nominal values.

3.9 Reproducibility

All figures and tables in Chapters 4 and 5 are generated by scripts in the project repository from mission parameter registers. The equation register documents every equation implemented in code with its source and units, and the parameter register carries a citation or assumed tag for every value. The validation suite of Section 3.7 runs on every change; regenerating the thesis results therefore only requires running the result scripts again. The simulator, equation register and validation suite are in the repository accompanying this dissertation, submitted with the project files. This thesis does not describe the graphical layer used during development, which is not necessary for regenerating the results.

3.10 Summary

The simulator is a chain of modules that each return a linear value, built from published equations with every unpublished value marked assumed and carried as a range, and checked by reproduction and measured comparisons that distinguish evidence from arithmetic. The choices that matter most for the results are Liang's clear-air model with DSOC's own Palomar allocation, the turbulence chain of [30], coding treated as a measured gap to capacity, and bisection on the forward model for the design questions. Chapter 4 reports the checks and the applications.

4 RESULTS AND ANALYSIS

This chapter presents the results produced by the simulator and assesses them against the five objectives set out in Section 1.2. Objectives 1 and 2 are addressed by the validation of Section 4.2, Objective 3 by Sections 4.3, 4.5 and 4.6, Objective 4 by Section 4.7 and Objective 5 by Section 4.8; the inverse-design validation of Section 4.4 underpins the Surrey study. Section 4.1 first sets out the validation scenarios and the assumptions which apply to all of them. Section 4.2 then presents the forward validation against published data, and Section 4.3 discusses which of those comparisons carry real weight, a question which needed more discussion than was originally planned (Section 3.7.1). Section 4.4 validates the inverse solvers and Section 4.5 then applies them to the Surrey ground station. Section 4.6 gives the sensitivity analysis, Section 4.7 discusses the coherent modulation study for Objective 4, and Section 4.8 solves the sized terminal for Objective 5. Finally, Section 4.9 brings the results together; the limitations of the work are stated in Section 5.3.

4.1 Validation Scenarios and Assumptions

The tolerance on each check is set individually, with the tolerance value following from the type of check it is. The two kinds of check are defined in Section 3.7, and the kind of each scenario family is given in Table 4.1.

Firstly, a reproduction check verifies that a stated equation from a published paper has been implemented correctly in this simulator, and is only limited by the printed precision on the paper itself. Where a paper gives a value that the model recomputes exactly, the tolerance can be very tight; the tightest decibel tolerances in the suite are 0.005 dB, on the atmospheric losses tabulated by [5]. Where the source prints fewer digits the tolerance follows suit. For example the Mars telemetry margin is published as 2.09 dB and is therefore checked to 0.05 dB, and the Klein and Degnan efficiency factor is published as 0.8145 and is checked to 5×10^{-4}. This validation method was helpful in catching modelling errors during production of the codebase, as discussed in Section 3.7.

Secondly, a measured check compares the simulator with measurements that were not used to build it, such as flight data from DSOC and GOPEX and ground-station measurements of turbulence. They are the looser of the two types, because the tolerance has to allow for the spread in the

measurements, which comes both from real changes in the atmosphere between passes or nights and from the uncertainty of each measurement. Where the source publishes that spread, it is used directly as the tolerance. For example, the Rytov variance at Table Mountain is checked against the measured 0.063 ± 0.02 [60], and the model gives 0.063. The width of DSOC's downlink beam, measured in flight, fell mostly between 13 and 15 μrad, and the model gives 14.2 μrad from the transmit telescope and its truncation ratio measured before launch [3].

Where no spread is published, the tolerance is wider and based on judgement. For example, the GOPEX flight comparison is held to 2.5 dB, a tolerance set by judgement when the check was written, because the measurements carry night-to-night changes in turbulence that the source does not quantify [34].

Several measured checks compare a ratio rather than an absolute value, so that parameters the source did not publish cancel out. GOPEX is compared relative to its first night, which removes the unpublished camera constants, and DSOC's rates at 0.56 and 2.68 AU are compared as a ratio, which largely removes the receiver-chain efficiency because it scales both rates together.

Other measured checks test a trend rather than a number. As an example, the aperture-averaging factor must weaken toward the horizon, as measured on the KIODO downlinks [48]; the simulator, using the form given by [36], reproduces this trend (Section 3.3).

Table 4.1: Validation scenario families. Kind: R = reproduction of a published calculation, M = comparison against measurement. Appendix A gives every check individually, with the quantity compared, its tolerance and result.

Scenario	Source	Kind
Transmit antenna gain	[6]	R
Inter-satellite link budget, LEO–LEO	[5]	R
Ground-satellite budget, LEO	[5]	R
Mars telemetry budget, 2.68 AU at 1064 nm	[10]	R
GOPEX laser uplink, Earth to Galileo	[34]	R, M
LLCD lunar downlink	[8]	M
DSOC deep-space downlink, 0.37–2.68 AU	[3]	M
DSOC link budget, 2.29 AU at 4 W	[3]	R
Atmospheric turbulence, HV$5/7$	[30, 61, 48, 60]	R, M
Background light	[40]	R
Modulation and coding	[4, 10, 42, 43]	R
Terminal register	[52]	R

Parameters that the sources do not publish are marked ASSUMED, and Chapter 3 describes how their ranges were chosen. One parameter that appeared to be unpublished was recovered from DSOC's own link budget. The receiver-chain efficiency, the fraction of the light collected by the

5 m Hale telescope that is detected, is not stated directly, but Table III of [3] lists each ground loss in decibels: optical transmission (-2.60 dB), other losses including jitter (-0.8 dB) and detection efficiency (-1.43 dB). Adding the -0.45 dB lost to the telescope's central obscuration, which the model's unobscured aperture does not include, gives a total of -5.28 dB, an efficiency of 0.30. Applying the table's favourable and adverse tolerances in the same way gives 0.38 and 0.25, so this parameter is carried as a published range rather than an assumed one throughout the chapter.

4.2 Forward Validation Against Published Data

This section presents the checks in two groups. The reproductions of published calculations (Section 4.2.1) test whether the equations were implemented correctly, and the comparisons with flight and ground-station measurements (Section 4.2.2) test whether the model describes real links. Section 4.2.3 records what the checks caught during development.

4.2.1 Reproduction of Published Calculations

All 65 checks pass: 48 reproductions of published calculations and 17 comparisons with measurement. The reproductions confirm that the equations are implemented correctly. The simulator reproduces DSOC's own downlink budget at its tabulated operating point of 2.29 AU and 4 W. DSOC's budget predicts a received power of -118.29 dBW and the simulator gives -118.32 dBW, a difference of 0.03 dB. The simulator takes DSOC's transmit optics and receiver-chain efficiencies from the same table but calculates the transmit gain, pointing loss and space loss itself, so the agreement shows that the budget is assembled correctly.

The simulator also reproduces the Mars telemetry budget of [10] (their Table III: 64-PPM, rate-1/3 SCPPM, 1064 nm, 2.68 AU) end to end to within 0.01 dB, giving a link margin of 2.08 dB against the published 2.09 dB. Finally, it reproduces the inter-satellite and ground-station link budgets of [5] to within 0.005 dB in every tabulated loss and transmit power, apart from the free-space loss at 5000 km in the inter-satellite budget.

Table 2 of [5] prints this loss as -272.18 dB, but scaling the 1000 km value of -258.18 dB to five times the distance gives -272.16 dB, which the simulator reproduces. The paper's own required transmit power of 29.30 dBm at that distance is also consistent only with -272.16 dB. The 0.02 dB difference is therefore a rounding slip in the published table, and that row is checked to 0.03 dB rather than 0.01 dB. Table 4.2 sets the published values against the simulator's, source by source.

Table 4.2: Validation: published link-budget values reproduced by the simulator, in order along the received-power chain. Difference is simulator minus published. The coherent-receiver reproductions are in Table 4.4; Appendix A lists all 48.

Quantity	Source	Published	Simulator	Difference
Transmitter and pointing				
Pointing loss, both terminals	[10], Tab. III	−8.45 dB	−8.45 dB	0 dB
Channel				
Space loss, Earth to Mars at 2.68 AU	[10], Tab. III	−373.49 dB	−373.51 dB	−0.02 dB
Clear-air (Mie) loss, 10° elevation	[5], Tab. 5	−0.57 dB	−0.57 dB	0 dB
Fried parameter, HV5/7, 0.5 μm, zenith	[31]	5.00 cm	4.96 cm	−0.04 cm
Detected background, horizon sky, 0.1 m^2	[40], Tab. 2	50.00 nW	50.27 nW	+0.27 nW
Received signal				
Received power, Mars telemetry link	[10], Tab. III	−130.96 dBW	−130.97 dBW	−0.01 dB
Received photon flux, Mars telemetry link	[10], Tab. III	−33.67 dB(pe/ns)	−33.68 dB(pe/ns)	−0.01 dB
Received power, DSOC, 2.29 AU, 4 W	[3], Tab. III	−118.29 dBW	−118.32 dBW	−0.03 dB
Capacity and coded rate				
SCPPM data rate, 64-PPM, rate 1/3	[10], Tabs. I–II	97.00 kb/s	97.00 kb/s	0 kb/s
Link margin and uplink				
Link margin, Mars telemetry link	[10], Tab. III	2.09 dB	2.08 dB	−0.01 dB
GOPEX signal on day 6 relative to day 1, against the experiment's theory	[34], Tab. 2	−11.88 dB	−12.47 dB	−0.59 dB

4.2.2 Comparison With Flight Measurements

GOPEX gives the first measured test, comparing the detected signal per laser pulse recorded at Galileo across a range of distances from 0.7 to 6.0 million km. Using only the published beam, range, seeing and zenith-angle parameters and a published clear-air transmittance [36], this model follows the experiment's own theoretical predictions to within 0.6 dB on every night [34], and three of the five nights agree with the flight measurements to within 0.4 dB. The experimenters summarised each night's pulses with a lognormal distribution, because turbulence caused the signal to vary from pulse to pulse [34]. The comparison uses the mean of that distribution rather than its median, because a link budget predicts the average received signal. This choice matters because the spread of the pulses changed from night to night, and using the median would have shifted the night-to-night comparison by up to 1.2 dB.

Day 2 falls 1.2 dB below the model, 1.7 times the standard error of that night's measurement, which reflects the uncertainty in an average taken over a limited number of pulses. Day 7 lies 2.2 dB above it, 2.6 times its standard error. Day 7 is the night on which the experimenters report a bimodal histogram of pulse energies, which they take as a sign that conditions such as turbulence or weather changed during the transmission, although they do not identify the cause, the seeing on each night was itself only roughly estimated [34]. Figure 4.1 overlays the model on the flight data; the value for each day is given in Appendix A (checks 53–62).

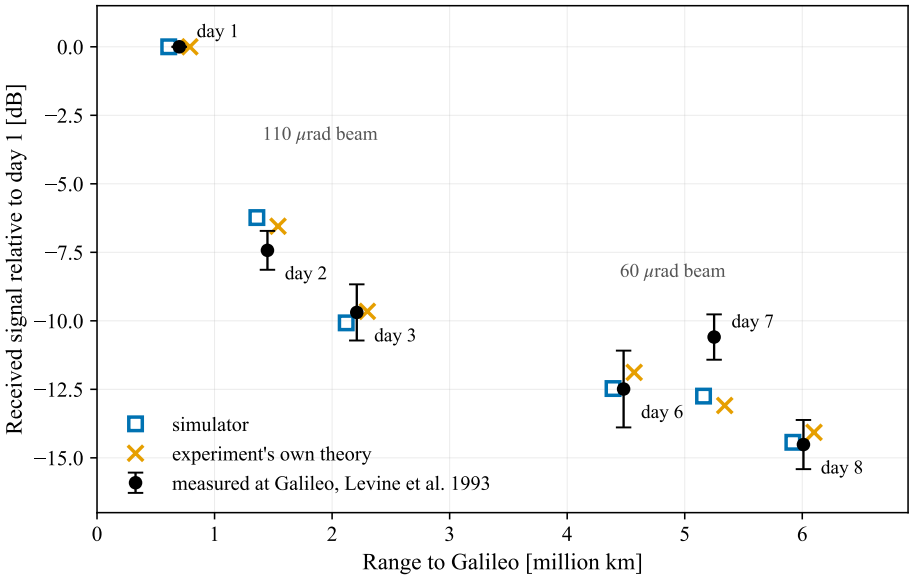


Figure 4.1: Validation: GOPEX 532 nm uplink to Galileo, December 1992, received signal relative to day 1 against range. Points: measured means with ± 1 standard error [34]; squares: the simulator; crosses: the experiment's own theory. The beam was narrowed from 110 to 60 μrad after day 3.

The turbulence model was tested on its own, because GOPEX cannot separate scintillation from the other night-to-night changes. Figure 4.2 compares the fading loss budgeted at 1% outage with measurements from the DLR optical ground station at Oberpfaffenhofen, which received LEO downlinks at 1550 nm through a 40 cm telescope, the same aperture as the Surrey station [61]. At 10° elevation the model gives 5.0 dB against a measured median of 4.2 dB (variability of 2.4–6.0 dB), and at 20° it gives 2.4 dB against a measured median of 1.9 dB (variability of 1.0–2.8 dB) [61], so both lie inside the measured spread.

The dashed curve in the figure is the model published by the operators of the same station [62]. Most of the difference between the two models comes from the turbulence models utilised. Their profile assumes a high-altitude wind of 10 m/s instead of the 21 m/s of HV5/7, and when this

simulator is run with 10 m/s, it gives 3.3 dB at 10°. The choice of profile is therefore worth more than a decibel at low elevation.

The HV5/7 turbulence profile also agrees with measurements at mountain sites. The modelled Rytov variance at Table Mountain, a measure of scintillation strength, is 0.063 against a measured 0.063 ± 0.02 [60]. The Fried parameter, the diameter over which the atmosphere stays coherent, is 5.0 cm at 500 nm, inside the ranges measured at Table Mountain (3.6–9.0 cm) [60] and Palomar (3.6–7.4 cm) [3].

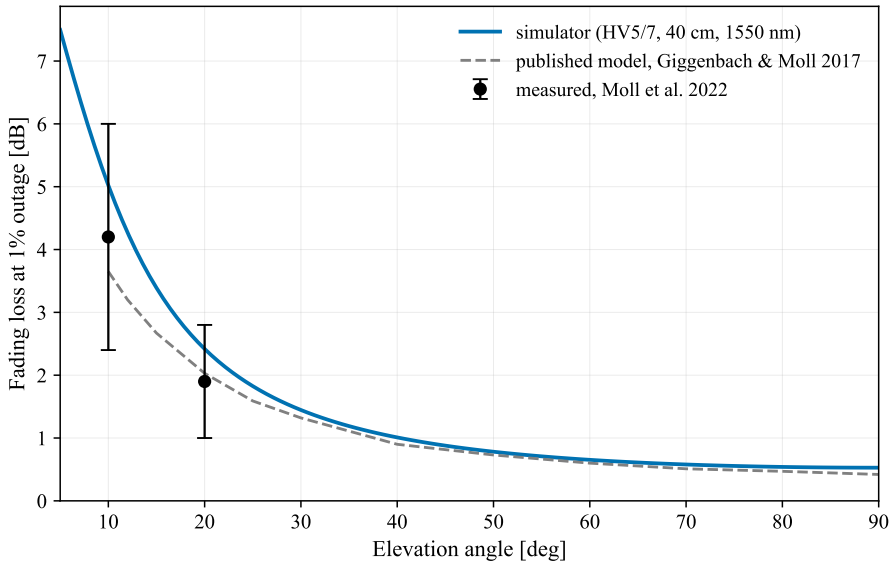


Figure 4.2: Validation: fading loss at 1% outage against elevation for a 40 cm aperture at 1550 nm (HV5/7). Points: measured median and spread at 10° and 20° [61]; dashed: the station operators' model [62], digitised from their Fig. 10.

The remaining measured checks are the mission benchmarks. The DSOC decode threshold is matched to within 0.25–1.6 dB of the flight-derived value across the published background range, and the LLCD link closes at its published rate. Appendix A lists all 17 measured checks, with the simulator's output against each published value. These checks do not carry equal weight, and Section 4.3 separates the comparisons that test the model independently from those that cannot fail.

4.2.3 What the Checks Caught

The two categories caught several modelling errors while the codebase was being built, and each category found a different kind of error. A reproduction check exposed a misprint in the source of the turbulence profile: [30] prints the ground-layer scale height of the Hufnagel-Valley profile

as 1000 m, which gives a Fried parameter of 1.44 cm at 500 nm instead of the 5 cm that defines HV$5/7$, whereas the standard 100 m form reproduces it. A comparison of the Gaussian and Airy-disk pointing models exposed a factor of four in the small-angle Airy approximation of the pointing loss. The two models disagreed by about 6 dB, which had been put down to the choice of model, and once the factor was restored all three models (Gaussian, exact Airy and its small-angle approximation) agreed at small angles. Measured checks exposed the two errors in the turbulence chain described in Section 3.3, through a fading loss of 0.7 dB at 10° against 4.2 dB measured [61], and a Fried parameter of 18–20 cm at mountain sites against 2.7–7.4 cm measured [3]. A reproduction can also expose an error in the source rather than in the model, as with the rounding slip in Table 2 of [5]. These errors also showed that a passing check does not prove a correct model. An early comparison of the OPALS link margin passed only because two errors cancelled: the transmit gain was 3.01 dB too low, from using the coefficient $c = 16$ discussed in Chapter 2 with a $1/e^2$ divergence, and the pointing loss was about 4.8 dB too severe. Section 4.3 takes up this question.

4.3 What the Validation Does and Does Not Establish

Applying the test of Section 3.7.1 to the flown missions, using what each source published (Table 2.4), separates the comparisons that test the rate model from those that cannot fail. Of the communication links checked against flight data, only DSOC is power-limited, at two of its three flight points, so it carries almost all of the weight of the rate validation, and the model passes both of those tests, as the next subsection shows. LLCD is a closure check: it confirms that the predicted power arrives at the receiver, but it does not test the rate model. OPALS and TESAT are modelled as reference configurations and have no measured check. Figure 2.4 in Chapter 2 shows all the flown missions from their published values.

4.3.1 DSOC / Psyche

DSOC provides the strongest test, because at two of its flight points the received photons set the data rate, so a prediction of that rate depends on every term in the link budget, and because its paper publishes both an itemised link budget with tolerances and flown rates at several distances.

Table 4.3 compares the rates DSOC flew with the simulator's prediction at each flight distance [3]. At 0.56 and 2.68 AU the link is power-limited, so the predicted rate depends on the whole received-power chain. At both distances the flown rate lies inside the range the simulator gives

across DSOC's efficiency tolerance, with the design value 25% above the flown rate at 0.56 AU and 13% below it at 2.68 AU. The range is wide at 2.68 AU because the simulated rate moves in steps between discrete signalling configurations. Between the two distances the flown rate fell by a factor of 16, and the model's fall, a factor of 14 to 23 across the efficiency range, contains it. The close match at 0.37 AU does not test the rate model, because that rate sits at the fastest configuration in the set.

Table 4.3: Validation: DSOC flown and simulated data rates. Simulator rates are at the design receiver-chain efficiency of 0.30, with the range across efficiencies of 0.25 to 0.38 in brackets. [a]The fastest configuration in DSOC's signalling set, reached at every efficiency.

Distance	Flown rate	Simulator rate
0.37 AU	267 Mb/s	266 Mb/s[a]
0.56 AU	133 Mb/s	166 Mb/s (132–199)
2.68 AU	8.33 Mb/s	7.27 Mb/s (7.24–14.5)

At the 20.6 Mb/s operating point of DSOC's link budget, at 2.29 AU and 4 W, DSOC's published budget gives 0.93 detected signal counts per symbol, with a standard deviation of about 0.1 counts, and the paper reports that the flight measurement matched this value [3]. The simulator is tested against this operating point in two ways. First, its received-power chain delivers 0.92 counts per symbol, calculated from the published transmitter, telescope and receiver-chain values. Second, it calculates the decode threshold, which is the number of detected signal photons per symbol at which the coded link just decodes. At the nominal background this threshold is 0.78 counts per symbol, 0.75 dB below DSOC's published operating point (about 0.7 dB across Monte Carlo seeds, Section 4.3.5), so the simulator is slightly optimistic, although part of the difference may be margin held above threshold in DSOC's operating point. Because the threshold is counted in detected photons, this second test checks the coding and detection model on its own, without depending on the receiver-chain efficiency. Comparing the two simulator values, 0.92 counts arriving against 0.78 needed, gives a predicted margin of 0.7 dB at this point, consistent with DSOC having decoded the link in flight [3].

Figure 4.3 shows the rate comparison against distance at the design efficiency of 0.30, over the 16- to 64-PPM orders and 0.5–2 ns slots DSOC flew, extended by an ASSUMED 128-PPM order and 4–8 ns slots for the weakest links. The modelled rate is flat at close range, at the fastest configuration in the signalling set, and falls below it once the received photons set the rate. Across

the wider set of 24 rates that DSOC flew, read from Fig. 17(a) of [3], none lies more than 0.02 dB above the model at the optimistic end of the efficiency tolerance. Most lie below it, as would be expected for passes flown through thin cloud or at a conservative rate.

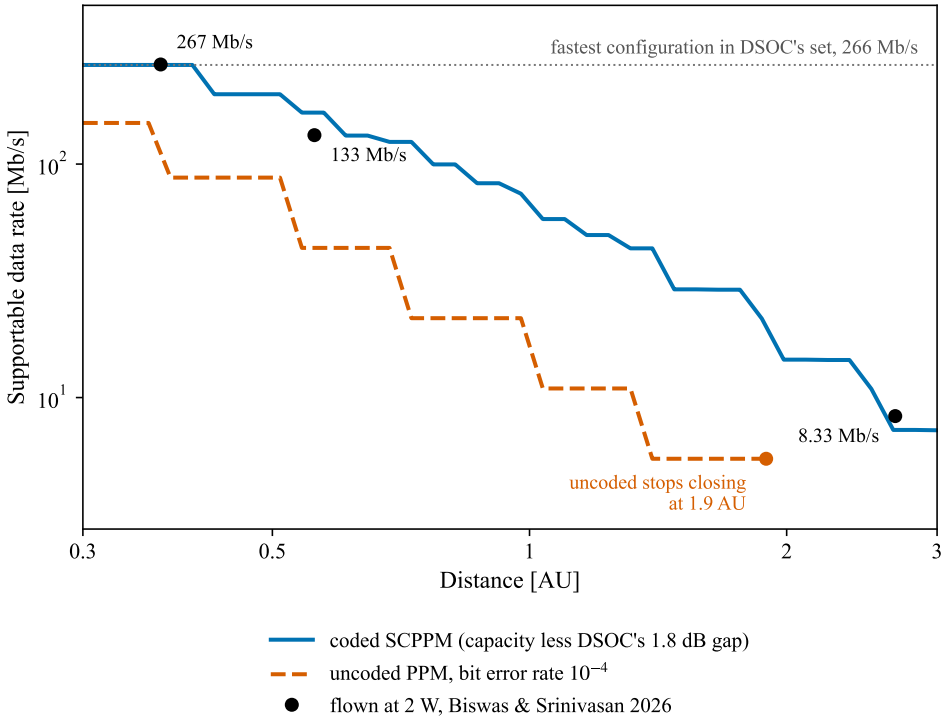


Figure 4.3: Validation: supportable data rate against distance for the DSOC-class downlink into the 5 m Hale telescope at 2 W, coded SCPPM (solid) and uncoded PPM at a 10^{-4} bit error rate (dashed) on a DSOC-class signalling set; dotted: the fastest configuration in the set. Points: the rates DSOC flew [3].

The dashed curve in Figure 4.3 shows what DSOC's code is worth on this link. Held to a 10^{-4} bit error rate on the same signalling set, uncoded PPM stops closing at 1.9 AU, meaning the flown measurement at 2.68 AU depended on the code. Where both close, the coded link carries 3 times the uncoded rate at 0.37 AU and nearly 7 times at 1 AU, and the advantage grows with distance as fewer photons arrive at the receiver.

In terms of hardware, matching the coded rate without coding would require DSOC's receiver to grow from the 5 m Hale telescope to about 9.8 m at 2.68 AU, twice the diameter and four times the collecting area. The code is therefore worth 5.9 dB at this distance, the same gain as that increase in telescope size, which is the penalty for uncoded links described in Section 3.6. Additionally, it is important to note that the figure compares a coded decoding threshold with an uncoded bit error rate, so it is an approximation rather than a like-for-like comparison.

4.3.2 LLCD

LLCD flew its downlink at 622 Mb/s from 0.0027 AU to MIT Lincoln Laboratory's ground terminal [8], and the simulator's budget closes at that rate. The simulator delivers 12.6 detected signal counts per symbol against its decode threshold of 1.22, a margin of 10.1 dB, although this rests on assumed optics efficiencies. No downlink margin is published for LLCD, so this confirms that the link closes rather than reproducing a published figure.

The reproduced data rate, however, is not evidence for the model's rate prediction's accuracy. The lunar link has power to spare, so the rate is set by its signalling format rather than by the budget, and reproducing it tests only that format. This is how the problem described in Section 3.7.1 was first found, when the published rate survived an artificial 13 dB penalty in the budget. LLCD is therefore counted as a closure check only, and the same reasoning was applied to the other missions.

4.3.3 OPALS

OPALS publishes a link budget with a current-best-estimate margin of 7.90 dB [47], the one published margin in the benchmark set. Its Table 1 states both a zenith angle of 65° and a range of 700 km, which cannot hold together for the ISS at 400 km: Eq. (2.5) gives about 840 km at 65° zenith, a difference worth about 1.6 dB of free-space loss. This is the range-elevation coupling of Section 2.3, and it is why a near-Earth budget cannot be reproduced from a stated range and a stated elevation that disagree. An earlier build of the simulator that carried an OPALS configuration sat 2.61 dB from the published margin, of which this inconsistency accounts for about 1.6 dB; the mission's own flight paper reports that the downlink power was reconciled only within the uncertainty of the atmospheric losses [21]. OPALS is therefore carried as a reference configuration and not as a measured check (Table 2.4).

4.3.4 TESAT LCT

The TESAT laser communication terminal [17, 18] is the flown example of homodyne BPSK, demonstrating inter-satellite links at 5.6 Gbit/s over distances of up to 5100 km between TerraSAR-X and NFIRE [17, 18]. Its crosslink budget is not reproduced in the validation suite, because TESAT publishes its data rate, modulation and bit error rate but not the link budget or receiver sensitivity that a reproduction would need (Table 2.4). The coherent receiver model is checked against two published results: the number of detected photons per bit that homodyne and hetero-

dyne BPSK need in theory [42], and how laser phase noise raises the signal needed by DPSK and sets the slowest usable data rate [43]. Every value produced by the model in Table 4.4 agrees to within 5% of its published value. The inter-satellite link budget is checked against the budgets of [5], reproduced in Section 4.2.

Table 4.4: Validation: the coherent receiver model against published values. The published 9 photons per bit is a rounded value, and the DPSK signal-to-noise ratio is read from a figure to about ± 0.5 dB.

Check	Source	Published	Simulator
Homodyne BPSK, photons per bit at 10^{-5}	[42]	4.55	4.547
Heterodyne BPSK, photons per bit at 10^{-5}	[42]	9	9.095
DPSK signal-to-noise ratio at 10^{-9}, 0.2 rad phase error	[43], Fig. 3	16.1 dB	16.03 dB
DPSK rate needed by a 10 MHz laser pair	[43]	3 Gbit/s	3.14 Gbit/s

4.3.5 What the Measured Checks Can Distinguish

The test of Section 3.7.1 can be applied more strictly by asking, for each measured check, whether a deliberately simple model would also pass it. Reading the checks this way, after the code had been frozen for submission, gave the following results. A model that carries only the free-space term and the nominal beam width, with no seeing and no atmosphere, reproduces the night-to-night GOPEX signal to within 1.6 dB on every night, so it passes all five flight checks at their 2.5 dB tolerance; its worst night is in fact closer than the simulator's own worst night (day 7, 2.2 dB). The same simple model misses the experiment's own theory by 1.0, 0.9 and 0.7 dB on days 6, 7 and 8 and so fails three of the five theory checks at 0.6 dB. It is therefore the comparison against the experiment's theory, not the comparison against the measurements, that tests the seeing and clear-air terms. For the DSOC rate ratio, a photon-limited rate falls as $1/d^2$, and $20\log_{10}(2.68/0.56) = 13.60$ dB; the simulator gives 13.59 dB at the design efficiency, so this check cannot distinguish the full receiver and coding model from the inverse-square law, and the flown 12.03 dB passes only because of the 2.5 dB tolerance. The counts-per-symbol check (check 42 of Appendix A) is the received power of check 41 expressed in detected photons over one symbol, so it repeats that reproduction rather than adding a measurement.

Two further points concern the decode threshold of Section 4.3. The 1.8 dB gap contains 0.8 dB of flight impairment taken from DSOC's own flight analysis ([3], Figs. 14 and 15); without it the modelled threshold sits 1.56 dB below the operating point rather than 0.75 dB, so part of the agreement is inherited from the mission's data. The Monte Carlo capacity uses a fixed random

seed; over four seeds the threshold moves between 0.66 and 0.75 dB below the operating point, so the figure is better quoted as about 0.7 dB.

Applying this reading to the whole suite, the checks that a trivial model could not pass, and that compare against a value not used in building the model, number about fifty of the 65. The fourteen that do not are the five GOPEX flight checks, the three DSOC ratio checks, the counts-per-symbol check, the two LLCD checks, whose tolerances of 0–15 dB and ± 4 dB accept any plausible budget, the ground-level C_n^2 check, which compares two constants with themselves, and the two NODE totals, which sum a published table. These findings were made after the code was frozen (Section 5.4), so the suite is reported as run, with all 65 checks, and the removal or replacement of the checks that add no evidence is left to future work (Section 5.5). They sharpen rather than weaken the conclusion of this section: the evidence for the rate model rests on DSOC's two power-limited flight points and on the decode threshold, and the evidence for the atmospheric terms rests on the GOPEX theory comparison and the fading measurements of Section 4.2.2.

4.4 Inverse Design Validation

The inverse solvers of Section 3.6 were validated in two stages. The first is a round trip (Figure 4.4): each solver's answer is passed back through the forward model it inverts and compared with the target it was given. For the DPSK rate solver, with the stabilised 100 Hz laser pair at 2.68 AU, the solved rate of 0.80 Mb/s returns a bit error rate of 3.800×10^{-3}, on the FEC threshold, while a rate 1% higher gives 3.99×10^{-3} and misses it. For the coded-PPM inversion, the 10 Mb/s uplink target requires a detected flux of 3.23×10^{6} photons/s. At that flux the forward rate search decodes 10.9 Mb/s, above the target only because the signalling configurations are discrete, and 1% less flux drops the rate to 8.31 Mb/s. Both solvers therefore stop at the edge of the forward model, neither short of it nor past it, which is what allows them to answer the design question that follows.

The second stage answers the question "what hardware would the University of Surrey need in order to send 10 Mb/s up to a 10 cm spacecraft telescope near Mars?" The solver's answer, using the site seeing as a limit floor on the uplink beam [34], is given in Table 4.5. The main result here is the trend across the rows. Beyond about 15 cm the required power ceases to fall steeply, because the seeing-limited beam, which is assumed to be 15 μrad at the site, no longer narrows as the aperture grows. This floor comes from the atmosphere itself, and the solver reproduced it without the behaviour being explicitly programmed in.

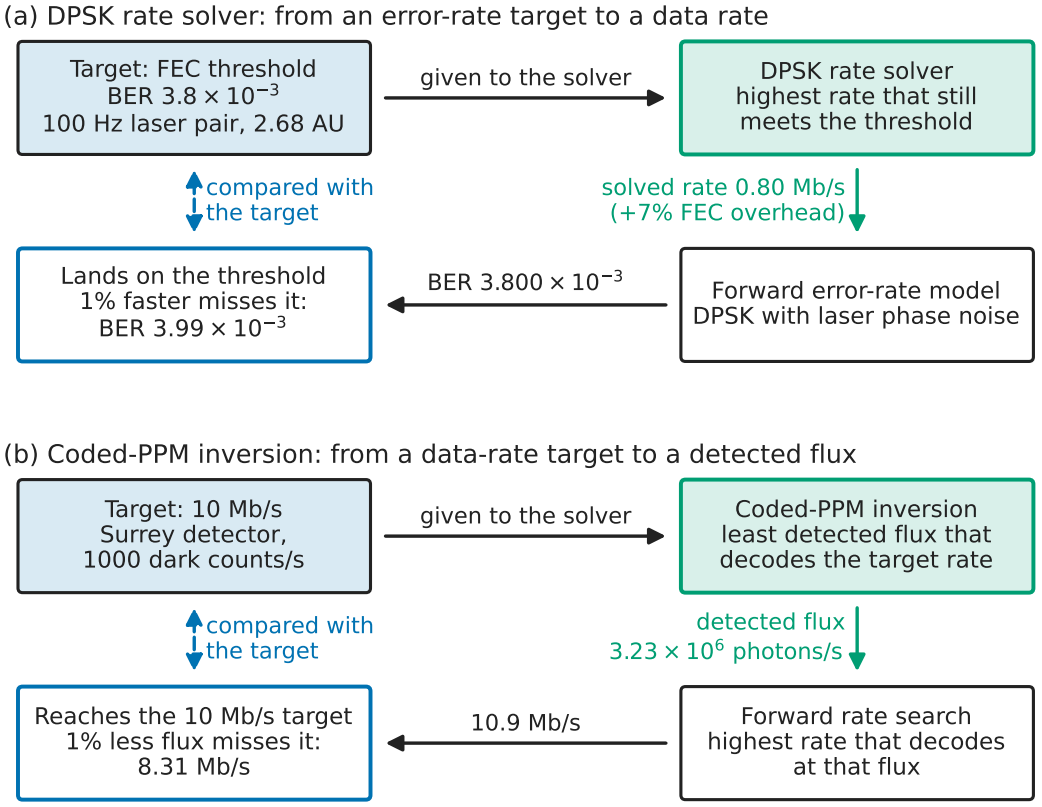


Figure 4.4: Simulator output: round trip of the two inverse solvers, (a) the DPSK rate solver at 2.68 AU with the 100 Hz laser pair and (b) the coded-PPM inversion for the 10 Mb/s uplink. Each answer is passed back through the forward model it inverts, and one step past it shows the answer sits on the boundary.

Table 4.5: Simulator output: laser power required at Surrey to deliver 10 Mb/s to a 10 cm spacecraft telescope at 0.37 AU, against ground aperture, with 15 μrad of seeing added to the diffraction width.

Ground aperture	Beam after seeing	Required power
5 cm	60 μrad	1399 W
15 cm	25 μrad	243 W
40 cm	17 μrad	119 W

The same solvers can also be used to draw the boundary of what the station can serve at all. Figure 4.5 fixes a 100 mW transmitter and asks what telescope closes a 1 Mb/s downlink into Surrey. At lunar range a few-millimetre optic is sufficient, whereas beyond about 0.49 AU no aperture closes the link at all, because the assumed 1 μrad pointing caps the transmit gain that a spacecraft can actually use. Past this boundary, adding aperture no longer helps and only more transmit power will close the link.

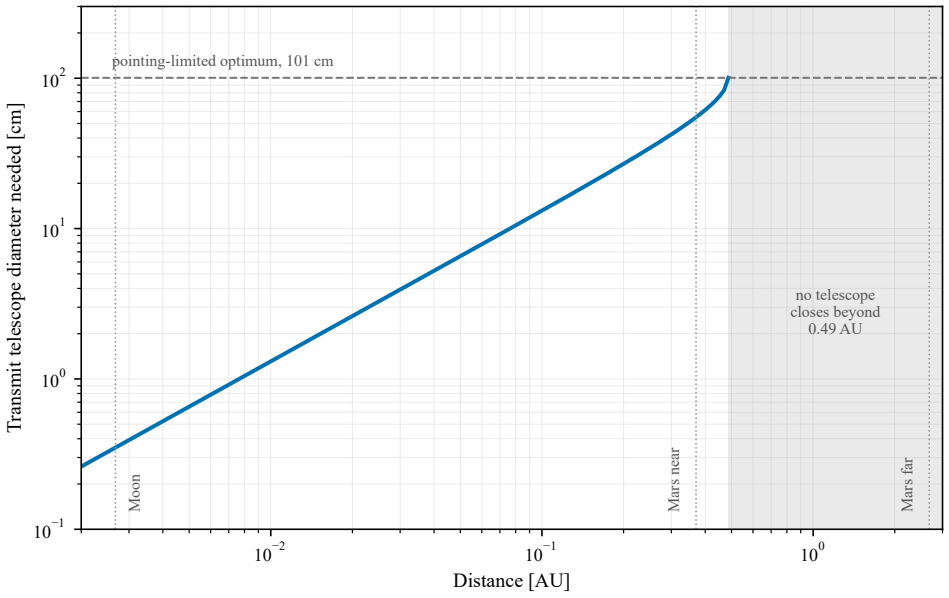


Figure 4.5: Simulator output: transmit telescope diameter needed to close a 1 Mb/s coded-PPM downlink into Surrey from a 100 mW laser, against distance, at 40° elevation on a clear night. Dashed: the pointing-limited optimum for 1 μrad; shaded: no telescope closes the link.

Repeating the exercise with the aperture fixed at 10 cm and the laser power left free tells the same story in different units. A lunar link closes on 0.1 mW, Mars at its nearest demands about 2.4 W, and Mars at its furthest about 120 W, so each factor of ten in distance costs a hundredfold in power, as expected from the $1/d^2$ term in the link equation.

4.5 A Deep-Space Link from Surrey

This section applies the model to a station the University actually operates, carrying the environmental and hardware realism of Objective 2 into a new case and comparing the outcome with DSOC's flight results (Objective 3). The University operates a 40 cm astronomical telescope [59], and the link budget model makes it possible to ask what that instrument could receive if it were pointed at a spacecraft's laser terminal instead of a star. Table 4.6 runs the station as an optical ground receiver at 40° elevation on a clear night, against a ladder of transmitter classes: a 10 Gb/s OOK crosslink-class terminal in LEO, an LLCD-class terminal at lunar range, and a DSOC-class terminal out to 2.68 AU, the farthest range at which Psyche's link was demonstrated. Station parameters, including the values assumed for its receiver, are listed in Table B.1.

A smaller telescope does not make the link fail at a fixed distance; it makes it slower. The 40 cm aperture collects about 160 times less light than the 5 m Hale telescope that received DSOC, yet

Table 4.6: Simulator output: Surrey 40 cm station as an optical downlink receiver at 40° elevation on a clear night. Transmitter classes from the cited missions and designs; station parameters as Table B.1. [a]The fastest configuration in the signalling set (4-PPM, rate 2/3, 0.5 ns slots), reached with power to spare, as for DSOC at 0.37 AU in Table 4.3.

Scenario	Transmitter class	Received power	Supportable rate
LEO, 600 km	10 Gb/s OOK crosslink-class	−5.1 dBm (margin +30.4 dB)	closes at 10 Gb/s
Moon	LLCD-class	−65.4 dBm	532 Mb/s coded PPM[a]
0.10 AU	DSOC-class	−80.8 dBm	58.5 Mb/s
0.50 AU	DSOC-class	−94.8 dBm	4.1 Mb/s
1.00 AU	DSOC-class	−100.8 dBm	1.0 Mb/s
2.68 AU	DSOC-class	−109.4 dBm	129 kb/s

a DSOC-class terminal at 2.68 AU would still deliver 129 kb/s into Guildford on a clear night. For comparison, DSOC returned 8.33 Mb/s through the Hale at the same range [3], although it transmitted 2 W there where this table assumes 4 W.

The rate holds up because a photon-counting PPM receiver can change how the data are packed into pulses. With the average laser power fixed, a longer PPM symbol puts more photons into each pulse, so a weak link can still be decoded if symbols are sent more slowly. At 2.68 AU the rate search chooses 256-PPM with slots of about 64 ns, far longer than any DSOC flew; restricted to a DSOC-class signalling set, the link into Surrey closes only out to 0.73 AU (Figure 4.6). At the calculated 129 kb/s from 2.68 AU range to Earth, an hour of contact between a DSOC-class laser terminal and the University of Surrey's telescope returns about 58 megabytes, enough for spacecraft telemetry but not for bulk science data. At this signalling each pulse carries about 80 μJ (4 W average over a 256-slot symbol and its guard time) and a peak power of order 1 kW; the model treats the transmitter as limited by average power only and does not check this against what a DSOC-class amplifier can deliver.

Figure 4.6 traces the supportable rate against distance for two receivers. The solid curve lets the rate search choose any PPM order and slot width, and the rate falls steadily with distance without reaching zero over the range plotted. The dashed curve is limited to a DSOC-class signalling set: it matches or sits just below the solid curve at short range, but it has no symbols long enough for a weaker link and stops closing at 0.73 AU. The freedom to slow the signalling is therefore what keeps a 40 cm station usable beyond that distance.

Figure 4.7 shows the link budget behind the solid curve as a trade between throughput and range. The transmitter is fixed as a DSOC-class terminal at 4 W, and the figure compares two quantities

at the Surrey station: the optical power the link delivers to the detector, after the 40 cm telescope and its relay optics, and the power the detector needs to decode each data rate once scintillation fading, the 0.5 detection efficiency and the dark counts are allowed for. The delivered power falls by 20 dB for every tenfold increase in distance, while the power needed rises by about 10 dB for every tenfold increase in data rate, since each bit needs a roughly fixed number of photons; each tenfold reduction in rate therefore buys roughly three times the range. Decoding 10 Mb/s needs −90.6 dBm (0.87 pW) at the detector, which the link delivers out to 0.31 AU, and decoding 1 Mb/s needs −101.1 dBm, which it delivers out to 1.0 AU. At 100 kb/s the link reaches 2.9 AU, just beyond Mars at its farthest, and at 10 kb/s it reaches 9.1 AU, close to the distance of Saturn. All other assumptions are those of Table B.1.

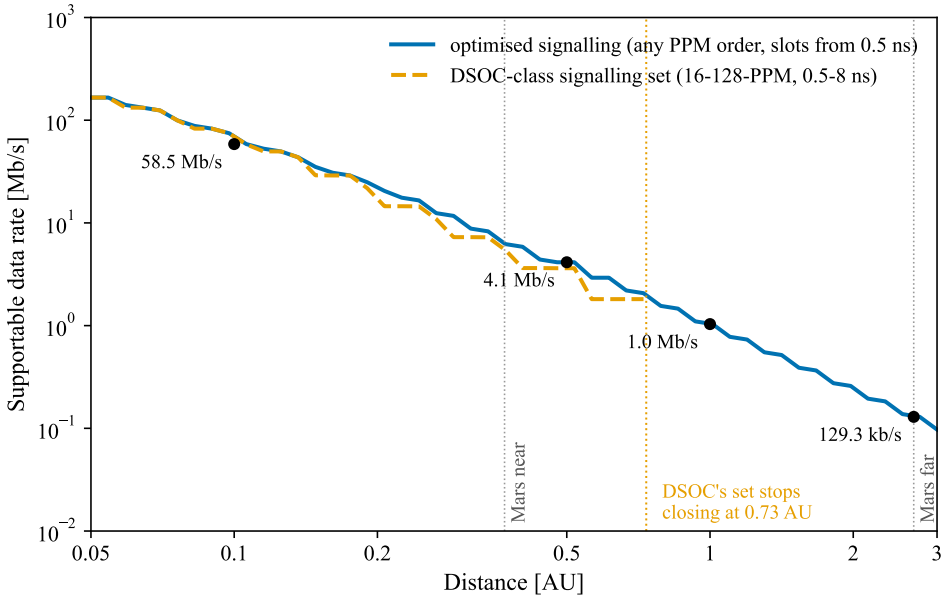


Figure 4.6: Simulator output: supportable coded-PPM rate into the Surrey 40 cm station from a DSOC-class 4 W terminal at 40° elevation on a clear night, with free choice of signalling (solid) and a DSOC-class signalling set (dashed). Points: Table 4.6.

Elevation matters more than the clear-sky air mass alone would suggest, because scintillation fading grows toward the horizon. For the DSOC-class link into Surrey, the supportable rate at 40° is 70–95% of its zenith value, and at 10° it is only 12–17%, as the fading loss budgeted at 1% outage rises from 0.5 dB at zenith to 5.0 dB. The exact fraction depends on range, because the rate moves in discrete steps. Low passes therefore remain usable, but at a substantial cost in rate.

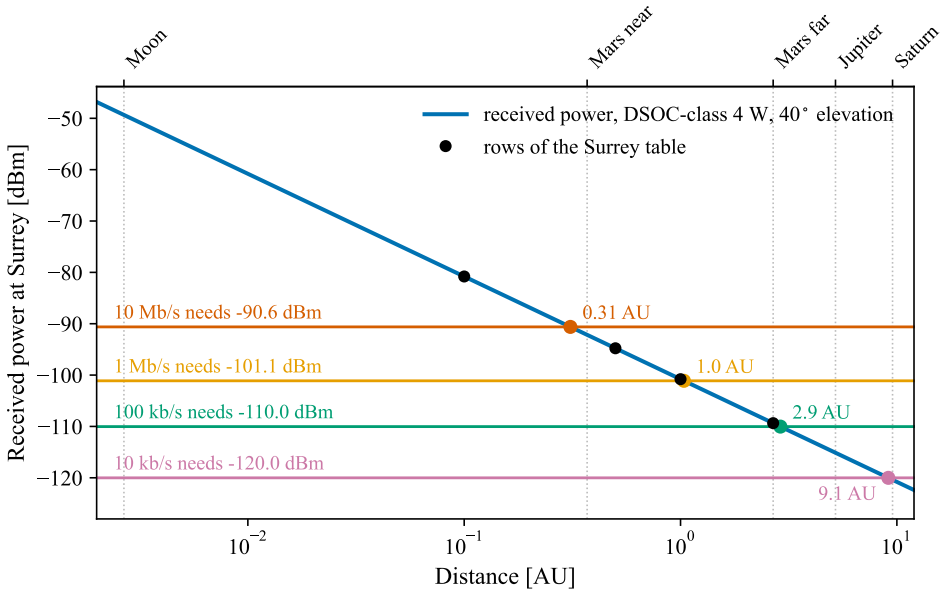


Figure 4.7: Simulator output: received power (falling line) and the power each coded-PPM rate needs (horizontal lines) for the DSOC-class 4 W downlink into Surrey at 40° elevation, against distance; dots mark the farthest distance each rate reaches. Points: Table 4.6.

4.5.1 What a Deep-Space Link into Surrey Would Take

The results of this section can also be read as a specification: what hardware would be needed for the University's 40 cm telescope to receive an optical downlink from a spacecraft in deep space. The specification is set at a single design point, a range of 2.68 AU (about 400 million km), which is the farthest range at which DSOC demonstrated its link and is close to the greatest distance between Earth and Mars. It is a distance rather than a Mars geometry: a spacecraft at Mars at that distance would lie almost behind the Sun as seen from Earth and could not be received on a dark night. At shorter ranges the transmit power needed for a given data rate falls with the square of the distance.

On the spacecraft side, a DSOC-class terminal (4 W laser, 22 cm transmit telescope, 1 μrad pointing) at 2.68 AU delivers 129 kb/s into the Surrey telescope on a clear night at 40° elevation (Table 4.6). Raising the Surrey link to 1 Mb/s at 2.68 AU needs the spacecraft to transmit about 124 W through a 10 cm telescope, or about 3 W through a 1 m telescope, which is the largest transmit aperture that 1 μrad pointing can use (Figure 4.5).

On the ground side, the 129 kb/s rests on a photon-counting receive chain behind the ODK16 with the assumed values of Table B.1: a telescope and relay optics throughput of 0.7 (0.5–0.85), a

detection efficiency of 0.5 (0.2–0.85) and 1000 dark counts per second (10^2-10^4). Taken to the pessimistic and optimistic ends of those intervals, the rate at 2.68 AU ranges from 32 to 275 kb/s. Moved one at a time, the detection efficiency changes it most, from 49 to 259 kb/s, and the optics throughput next, from 92 to 183 kb/s; on a dark night the dark-count rate, field of view, filter width and seeing do not change it at all. None of these values have been measured at Surrey. The optics throughput belongs to the existing telescope and could be measured directly, but the detection efficiency and dark-count rate describe a photon-counting detector that the University may not yet have. These two values are therefore best read as the requirements such a detector would have to meet, and the 129 kb/s as what the station could achieve with it, not as a prediction for the telescope as it stands today. The budget also carries no receive pointing loss, because the telescope is assumed to hold the spacecraft inside its 100 μrad field of view throughout a pass. Whether the ODK16 mount can track a deep-space target to that precision was not assessed, and it is the first thing a hardware study should establish.

Operationally, the 129 kb/s holds for a clear, dark night with the spacecraft at 40° elevation. Scintillation fading grows toward the horizon, so the same link supports 183 kb/s with the spacecraft overhead but only 23 kb/s at 10° elevation (Section 4.5). Cloud statistics for Guildford are not modelled in this simulator, so the fraction of time the link is available is unknown.

An uplink is a separate matter, because the station is a receiver and an uplink would need a laser transmitter added at the site. Sending 10 Mb/s from the site to a 10 cm spacecraft telescope at 2.68 AU would need about 6.3 kW of laser power even through a 40 cm transmit aperture, which is the 119 W of Table 4.5 at 0.37 AU scaled by the square of the distance. That requirement is set by the site seeing rather than by the aperture, and it is a lower bound, since the seeing is carried as a $1/e^2$ width and no uplink scintillation is modelled (Section 5.3).

Taken together, at 2.68 AU the spacecraft side of a Surrey downlink is within reach of a terminal that has already flown, provided a rate of about 100 kb/s is acceptable, and the ground side is a photon-counting receiver of stated efficiency and dark-count rate, on a mount whose tracking has yet to be checked. An uplink from the site at this range would need kilowatt-class laser power. Cost, availability and the receiver hardware itself are outside what this simulator can settle, and they are deferred to Chapter 5 as future work.

4.6 Sensitivity Analysis

With the model validated as far as the available data allow, the natural next question is which parameter is most worth improving, which is the parameter-sensitivity part of Objective 3. The analysis uses the DSOC downlink at 1 AU as flown: a 4 W transmitter with a 22 cm telescope, received by the 5 m Hale telescope, with a supportable rate of 124 Mb/s from DSOC's link budget. Each hardware parameter was made 10% worse and 10% better with the others held at their baseline values (Figure 4.8). Moving every parameter by the same fraction keeps the ranking independent of how wide a range each parameter is given. This is a one-at-a-time sweep and does not capture interactions between parameters, but it is adequate for an overall idea of the sensitivities.

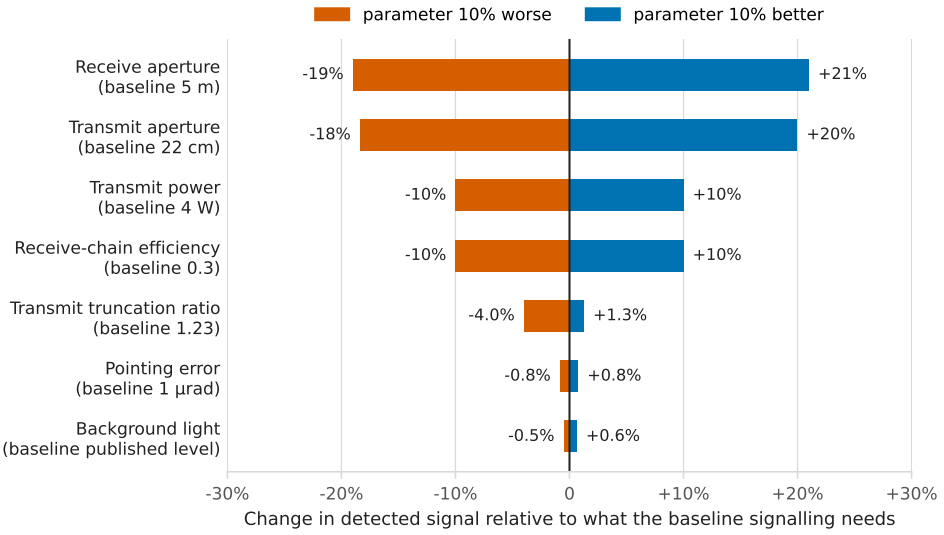


Figure 4.8: Simulator output: change in detected signal for the DSOC downlink at 1 AU (4 W, 22 cm, 5 m Hale, 124 Mb/s baseline) when each hardware parameter is made 10% worse (orange) and 10% better (blue), one at a time.

The apertures of both the transmitter and receiver were found to dominate following the sensitivity analysis. A 10% change in the diameter of either the receive or the transmit telescope changes the detected signal by about 20% (receive -19% / $+21\%$, transmit -18% / $+20\%$), because the collecting area of the receiver and the gain of the transmitter both scale with the square of the diameter. Transmit power and receiver-chain efficiency act linearly and change the signal by 10%. The transmit truncation ratio changes it by -4.0% / $+1.3\%$, because the baseline of 1.23 already sits close to the model's optimum of about 1.12. Pointing error changes it by less than 1%, because 1 μrad is small against the width of the beam, and background light also by less than 1%, because DSOC's measured background is only 0.003 counts per 1 ns slot [3], about 0.0015 at the 0.5 ns slots of the baseline configuration. For a terminal designer, a telescope 10% larger at either end

therefore buys about twice as much signal as 10% more laser power, which is consistent with DSOC's use of the 5 m Hale telescope as its receiver.

The same model bounds how much the DSOC comparison depends on its least certain parameter, the receiver-chain efficiency. DSOC's own link budget gives it a tolerance of 0.25 to 0.38 about the 0.30 baseline [3]. Across that tolerance the detected signal changes by -17% to $+27\%$, about the same as making the 5 m Hale telescope 9% smaller or 13% larger in diameter, and the supportable rate changes from 124 Mb/s to 99.7 Mb/s (-20%) or 132 Mb/s ($+6\%$). The same is not true of the Surrey station, whose receiver values are assumed (Table B.1): across their intervals the rate at 2.68 AU moves from 129 kb/s to between 32 kb/s (-75%) and 275 kb/s ($+113\%$). Section 4.5 therefore treats those values as a specification to procure against, not as a prediction.

4.7 Coherent Modulation with a Stabilised Source

Objective 4 asks what a narrow-linewidth source would enable. The analysis answers this in two steps: first, what coherent detection is worth, and second, what linewidth it requires.

On the first point, the shot-noise-limited sensitivities of Section 2.4 put homodyne BPSK at 4.55 detected photons per bit for a 10^{-5} error rate, against 9.0 for heterodyne, whereas noiseless 16-ary photon counting needs only 2.7 [42, 4]. Coherent detection therefore does not beat photon counting on sensitivity. Its advantages lie in spectral efficiency and daylight tolerance, as anticipated in Chapter 2.

On the second point, the combined linewidth of the transmitter laser and the local oscillator that DPSK can tolerate scales in proportion to the data rate [43], so it becomes very small at deep-space rates. Table 4.7 gives it at each rate, using the criterion that the residual phase error stays within 0.2 rad, the level at which the receiver-sensitivity loss stays below 3 dB [43]; the same relation reproduces that paper's own result that 10 MHz lasers need DPSK rates above 3 Gb/s. The 0.2 rad budget is taken from that paper's 10^{-9} error-rate curves and applied here at the 3.8×10^{-3} FEC threshold, where it is conservative. Standard distributed feedback (DFB) lasers, with linewidths from 100 kHz to 10 MHz [42], need at least 31 Mb/s and up to 3.1 Gb/s, which rules them out at the kb/s to few-Mb/s rates of links at Mars distances, whereas a 100 Hz stabilised pair supports every rate above about 31 kb/s. Coherent modulation at deep-space data rates is therefore limited by laser stability rather than by the modulation scheme, and becomes practical once a stabilised source is available, an option which this project initially planned to demonstrate.

Table 4.7: Simulator output: largest combined transmitter and local-oscillator linewidth that heterodyne DPSK tolerates at each data rate for a 0.2 rad residual phase error [43], and whether a matched pair of each laser class (twice the linewidth of one laser) meets it.

Data rate	Largest tolerable combined linewidth	Stabilised pair (100 Hz each)	DFB pair (100 kHz each)	DFB pair (10 MHz each)
10 kb/s	64 Hz	no	no	no
100 kb/s	637 Hz	yes	no	no
1 Mb/s	6.4 kHz	yes	no	no
10 Mb/s	64 kHz	yes	no	no
100 Mb/s	637 kHz	yes	yes	no
1 Gb/s	6.4 MHz	yes	yes	no
10 Gb/s	64 MHz	yes	yes	yes
Lowest workable rate	–	31 kb/s	31 Mb/s	3.1 Gb/s

The linewidth budget sets only the slowest usable rate. The fastest is set by the weakness of the signal: at the 3.8×10^{-3} FEC threshold, heterodyne DPSK needs about 5.2 detected photons per information bit, from the error-rate model of [43] (Eq. (36)) with the 7% FEC overhead included. Figure 4.9 combines the two limits for a DSOC-class downlink at 2 W, the power of the DSOC flight points shown [3]. The coherent receiver is assumed to detect the same photon flux as DSOC's photon-counting receive chain, with no loss to atmospheric turbulence. This is an upper bound: at a ground station, turbulence degrades heterodyne mixing unless adaptive optics or a space relay is used [31]. At a 5 m ground aperture with a Fried parameter of about 20–27 cm at 1550 nm (the 5–7 cm of Section 4.2 at 500 nm, scaled by $\lambda^{6/5}$) the heterodyne mixing efficiency falls roughly as $(r_0/D)^2$, about 25–30 dB, so the DPSK windows describe a space-relay or adaptive-optics receiver rather than a bare ground telescope. Each laser pair supports DPSK between the lowest rate its linewidth allows, drawn as that pair's horizontal line, and the photon limit. On this link a pair of 10 MHz DFB lasers loses its window beyond 0.04 AU, and a 100 kHz pair beyond 0.41 AU, just past Mars at its closest. The 100 Hz stabilised pair keeps a window open to about 13 AU, beyond Saturn, and at 2.68 AU it still spans 31 kb/s to 0.80 Mb/s. The distances at which the windows close are not properties of the lasers alone; they also depend on the link. With more transmit power or a larger telescope, more photons arrive, the fastest usable rate rises, and every window stays open to a greater distance. The slowest usable rate of each laser pair does not change, because it depends only on the linewidth of the lasers.

The figure also shows why photon counting remains the choice for deep space. Coded PPM on the same link overtakes the DPSK photon limit beyond about 0.10 AU, and delivers 7.1 times its rate at 0.37 AU and 15 times at 2.68 AU. The rates DSOC flew sit at 71–89% of the PPM line, as expected for a fixed signalling set rather than the free choice of PPM order and slot width that the

line assumes. A stabilised laser therefore makes coherent modulation possible at interplanetary distances, but it does not make it more efficient than photon counting there. The comparison does favour PPM on coding: the PPM curve is capacity less 1.8 dB, whereas DPSK is held to a hard-decision FEC threshold with 7% overhead, so a capacity-approaching soft-decision code for DPSK would narrow, though not close, the gap.

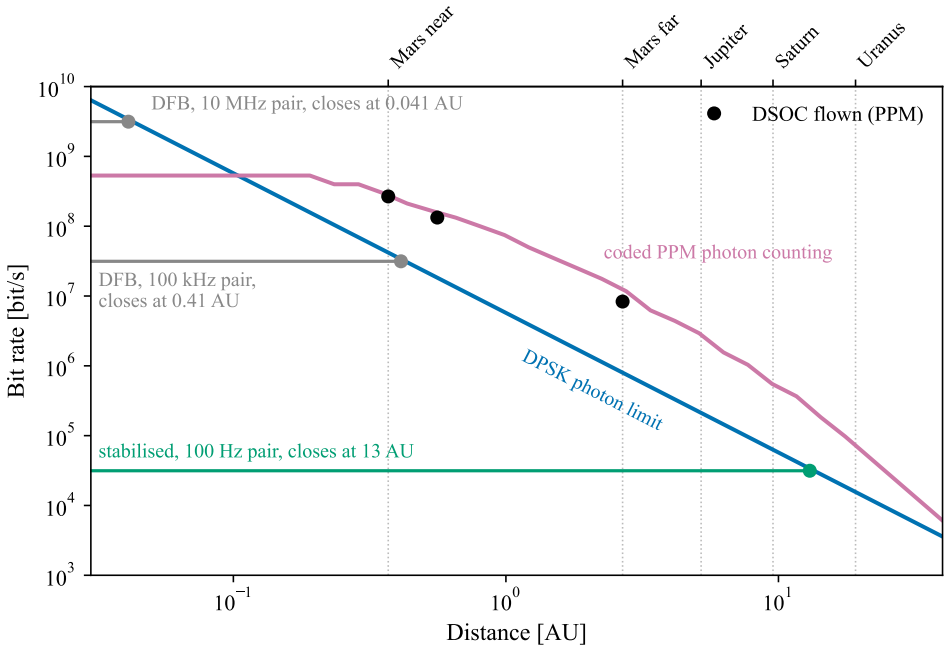


Figure 4.9: Simulator output: DPSK windows for three matched laser pairs and the coded-PPM rate (purple) on a DSOC-class downlink at 2 W, against distance. Each window runs from the slowest rate the pair's phase noise allows (0.2 rad) up to the DPSK photon limit at the FEC threshold; dots mark where it closes. Points: the rates DSOC flew [3]. Top axis: planetary distances [36].

4.8 System Sizing: A Representative Terminal

Objective 5 asks for a full component list for a representative CubeSat FSO terminal, and the register is built on the NODE terminal of Section 2.5, which is the most completely documented design at this scale [52]. Table 4.8 totals it by subsystem. The terminal comes to 550 g of a 600 g mass budget and 8.5 W of a 10 W power budget, and both totals check against the published tables. Each entry carries a provenance flag (measured, vendor or estimated), and unit costs are recorded as not available rather than guessed, because the source does not publish them and this project does not invent numbers.

The single largest component is the EDFA, the optical amplifier within the transmitter subsystem of Table 4.8. At 150 g and 4.1 W [52], it makes up most of that subsystem's 220 g and 4.6 W,

Table 4.8: Compiled from published values: representative CubeSat FSO terminal by subsystem, after NODE [52], with a provenance flag on every entry in the register. Costs unpublished, recorded as n/a.

Subsystem	Mass [g]	Average power [W]
Transmitter (seed laser, EDFA, optics)	220	4.60
Beacon receiver	150	2.00
Structure and optics bench	100	0.00
Electronics (FPGA, framer, buffer)	60	1.25
Fine pointing	20	0.50
Radio (TT&C modem)	0	0.18
Total	550	8.5

and accounts for 27% of the terminal's 550 g mass and 48% of its 8.52 W average power. This connects back to Section 4.6. Doubling the transmit power doubles the detected signal, a gain of 3 dB, which on the DSOC-class link at 1 AU raises the supportable rate by 60%, from 124 to 199 Mb/s. On this terminal, however, the 10 W power budget leaves only 1.48 W of headroom. If the amplifier's electrical draw scales with its optical output (ASSUMED), that headroom allows at most 36% more transmit power, about 1.3 dB, and doubling the output would take the terminal to about 12.6 W, over its budget. The same doubling of detected signal costs the spacecraft nothing if it is found on the ground instead, where it needs a receive telescope only 41% larger in diameter, for example 57 cm instead of Surrey's 40 cm. A small terminal's upgrades are therefore more easily found in the aperture of the ground segment than in the power supply of the spacecraft. The register is a compilation with provenance rather than a new design, and it is not yet coupled to the link model: sizing which terminal in this class closes a given link into Surrey, and what the aperture-for-power trade of Section 4.6 costs in mass, is the natural next step and is listed in Section 5.5.

4.9 Summary

This chapter tested the simulator against published calculations and measurements, used it in the inverse direction to answer design questions, and applied it to the Surrey ground station, to coherent modulation and to a representative terminal. What it established is summarised below; Chapter 5 evaluates the results against each objective.

Taken together, the results show several things beyond whether each objective was met. Agreement with a published value is not on its own evidence that a model is correct: only a power-limited link such as DSOC tests the rate prediction, because a bandwidth-limited link such as LLCD repro-

duces its published rate even with a 13 dB error in the budget. On deep-space links the telescope apertures matter more than any other hardware parameter, and coding is what makes the longest links possible, since DSOC's 8.33 Mb/s at 2.68 AU would otherwise have needed a receiver of about 9.8 m in place of the 5 m Hale. Physical limits other than power also shape the design: site seeing means an uplink telescope larger than about 15 cm gains little, and a 1 μrad pointing error means a spacecraft telescope larger than about 1 m gains nothing. For coherent modulation the limit at deep-space rates is laser stability rather than the modulation scheme, and even with a stabilised source photon counting delivers more data at Mars distances. Finally, a 40 cm university telescope could receive 129 kb/s from a DSOC-class transmitter at Mars's farthest distance on a clear night, which suggests that deep-space optical reception is not limited to large observatories, provided the receiver hardware described in Section 4.5.1 can be built and tested.

Chapter 5 draws these findings together into the conclusions of the project, reflects on how the objectives were delivered, and sets out the future work, including a statistical availability model for the Surrey site.

5 CONCLUSIONS

This project set out to develop an open, validated simulator for free-space optical satellite links and to apply it to real questions, from flown deep-space demonstrations to a single telescope in Guildford. This chapter summarises what was achieved, evaluates it against the objectives and the work of others, states the limitations of the work, reflects on how the project was carried out, and sets out what should follow.

5.1 Summary of Achievements

The project produced a link-budget simulator for free-space optical satellite links built only from published equations and parameters, with every formula traced to its source and every unpublished value marked as assumed. Its validation suite contains 65 checks, all of which pass: the 48 reproductions pass within their stated tolerances, and the published link budgets of DSOC, [10] and [5] are matched to within 0.03 dB. Additionally, the 17 comparisons with published measurements all pass within their stated tolerances: the model matches DSOC's measured beam width to within 1.2%, reproduces the ground-station turbulence measurements within their measured spread, and follows the GOPEX uplink to within 0.4 dB on three of its five nights, with a largest difference of 2.2 dB. Section 4.3.5 sets out which of these checks a trivial model would also pass.

The validation led to a test for separating evidence from arithmetic in mission benchmarks. It showed that only DSOC tests the rate model independently, and that the model passes at both of DSOC's power-limited flight points. The simulator was then used to produce a first specification for a deep-space optical downlink into the University's 40 cm telescope, together with the transmitter, receiver and operating conditions such a link would need. Finally, the project quantified the case for a stabilised laser, which keeps a DPSK window open to about 13 AU against 0.41 AU for a 100 kHz DFB pair, and gave a component breakdown of a representative CubeSat terminal.

5.2 Evaluation Against the Objectives

Objectives 1 to 4 were met as stated, although not always in the form that was anticipated, and Objective 4 changed considerably; Objective 5 was met for hardware, mass and power but not for cost, which no available source publishes. The baseline simulator (Objective 1) and the environmental and hardware extensions (Objective 2) are supported by the validation suite, the OOK

case being closed against a published receiver sensitivity rather than a detector noise model (Section 3.4). The turbulence comparison was the most eventful part of this work. It initially disagreed with the measured fading by about 3.5 dB at 10° elevation, and tracing this disagreement exposed two errors in a secondary source's statement of the equations (Section 3.3). Against the work of others, the corrected model gives 5.0 dB of fading at 10° for a 40 cm aperture, inside the spread of the ground-station measurements around their median of 4.2 dB, and lies within about 1.4 dB of the model published by the operators of the same station, a difference traced mainly to the wind speed assumed in the turbulence profile (Section 4.2).

Objective 3 was to benchmark the simulator against real missions and to explore its sensitivity to its parameters. It produced what is probably the most important result of the project, and one about method rather than numbers: agreement with a flown mission is not the same as validation. For LLCD the reproduced data rate was circular, because the model's slot clock had been derived from the same published rate, and OPALS and TESAT could only be modelled as reference configurations. This was not noticed immediately. Only DSOC, which is power-limited at two of its three flight points, tests the rate model independently. At those two points, the flown rates at 0.56 and 2.68 AU lie inside the range the model gives across the efficiency tolerance of DSOC's own link budget, and the modelled decode threshold lies about 0.7 dB below DSOC's published operating point (Section 4.3). That tolerance alone moves the detected signal by -17% to $+27\%$ (Section 4.6), so the model agrees with the flight data to within the uncertainty that the mission itself carries. The sensitivity analysis adds a design conclusion: a 10% change in the diameter of either telescope changes the detected signal by about 20%, twice the effect of the same change in transmit power.

Objective 4 was descoped from a hardware demonstration to an analytical study when the stabilised source could not be delivered to the Surrey Space Centre in time. The analysis gives the following answer: coherent modulation at deep-space data rates is limited by laser stability, and Chapter 4 gives the tolerable linewidth at each data rate. Standard DFB lasers, with linewidths of 100 kHz to 10 MHz, need DPSK rates above 31 Mb/s, and a 100 kHz pair loses its window beyond 0.41 AU, whereas a 100 Hz stabilised pair keeps one open to about 13 AU. The same analysis also showed the limit of the idea, since coded PPM still delivers 7.1 to 15 times the DPSK rate between 0.37 and 2.68 AU; a stabilised source makes coherent modulation possible in deep space without making it the better choice there (Section 4.7). Objective 5 finished with a component breakdown

of the NODE terminal at 550 g and 8.52 W, within its published mass and power budgets, with the optical amplifier alone taking 48% of the power. Component costs are not published, so they are marked as unavailable rather than estimated, and this objective is therefore met for mass and power but not for cost.

Beyond the objectives, the simulator was used to model a theoretical link between a DSOC-class transmitter at 2.68 AU and the University of Surrey's 40 cm telescope. On a clear night, with the transmitter and receiver switched to much slower signalling than DSOC flew (256-PPM with slots of 64 ns), the station would receive 129 kb/s, and between 32 and 275 kb/s across the assumed receiver values (Section 4.5). This result depends on parameters of the station that are not known: the aperture and secondary mirror of the telescope come from its specification, but the throughput of its optics, the detector the link needs, the site seeing and the tracking precision of the mount are all assumed or unmeasured. Because the rest of the link is also idealised, with a completely dark sky, perfect tracking and a modem using signalling far slower than any flown, the value is more likely to be optimistic than pessimistic (Section 4.5.1). Within those caveats, an instrument bought for astronomy appears to be within range of a deep-space downlink, and Section 4.5.1 sets out what closing such a link would take on the spacecraft, at the station and in operation.

5.3 Limitations

The limitations of the work fall into three groups: how far the validation reaches, what the model leaves out, and the assumptions behind its applications.

The validation of the rate prediction rests on a single mission. Only DSOC tests it independently, at two of its three flight points; LLCD confirms only that its link closes, and that on assumed optics efficiencies, while OPALS and TESAT have no measured check. This matters because one mission cannot show whether the agreement holds for other hardware, ranges or ground sites, so an error that happens to cancel for DSOC's configuration would go undetected. Additionally, the turbulence model uses the generic HV$5/7$ profile at every site and has been compared only with night-time measurements, each taken at a single site. Turbulence differs between sites and is stronger by day, so the fading predicted for another site, or for daytime, has not been tested, and the uncertainty is large: with a daytime ground-layer strength the simulator predicts 2.2 dB of fading at 10° for a 40 cm aperture, whereas the model of Giggenbach and Moll, applied to the same station, gives 7 dB [62], and none of the measurements used here can separate the two. Two choices in the turbulence equations also remain uncertain because the sources disagree. The simulator

measures the height in the Hufnagel-Valley profile from the local ground, as ITU-R P.1621-2 defines it, but the same document plots the profile and states its Fried-parameter approximation against height above sea level [37]. This leaves the convention open, and it matters most at elevated sites, where the two readings differ by the whole ground layer. The aperture-averaging coefficient likewise differs between sources, 1.11 in [30] against about 0.91 in [36]. The simulator uses 1.11, and the alternative would change the fading allowance for a 40 cm aperture by 0.1–0.4 dB. Some tolerances, such as the 2.5 dB on the GOPEX comparison, were set by judgement because the source does not publish the spread of its measurements, so a pass on those checks depends partly on that choice rather than on the data alone.

Next, the model is a static link budget: pointing error is held constant with range, so any growth in pointing error on the longest links is not captured and those results may be optimistic, although on the DSOC link a 10% change in pointing error moves the detected signal by less than 1%. Solar conjunction geometry, which interrupted DSOC's own operations [3], is not included, so the model cannot predict when a deep-space link will be lost as its target passes close to the Sun. The sensitivity analysis moves one hardware parameter at a time, so it cannot show whether two parameters that are each tolerable on their own become significant together, although the unmeasured Surrey station values are evaluated together at the pessimistic and optimistic edges of their ranges. Nothing in the model varies with time: cloud, season and the position of a target in the sky are not modelled. It therefore gives the rate a link supports under stated conditions but not how often those conditions occur, which is what would be needed to estimate how much data a station could return over a year.

Finally, the applications of the model also carry further assumptions. The Surrey results rest on station parameters that have not been measured, namely the optics throughput, the detector, the site seeing and the tracking of the mount, so they describe a station with those values rather than the telescope as it stands, and across the assumed ranges the rate at 2.68 AU spans 32 to 275 kb/s. The link around them is idealised, with a completely dark sky, perfect tracking, a photon-counting array free of dead time and signalling far slower than any flown, and its coding and synchronisation loss is DSOC's measured 1.8 dB gap to capacity [3], carried over to a receiver that has not been built. Each of these either removes a loss or assumes a capability that a real station does not yet have, so the 129 kb/s at 2.68 AU is more likely to be optimistic than pessimistic. The inverse-solved uplink includes the average spreading of the beam by turbulence, through the site seeing,

but not the fluctuations about that average, the scintillation and beam wander that make the signal at the spacecraft fade, because only a downlink fading model is implemented and validated; its required transmit powers are therefore lower bounds. The coherent study assumes a receiver that detects as many photons as DSOC's photon-counting chain with no loss to turbulence, whereas turbulence distorts the wavefront that coherent detection depends on, so its DPSK windows are upper bounds. The terminal breakdown lacks component costs, which no available source publishes, so Objective 5 is met for mass and power but not for cost.

5.4 Reflection on the Project

The project met the objectives set out in its original plan, but not in the planned order or with the planned emphasis. The plan treated the simulator as a sequence of layers, a baseline link budget, then atmospheric and hardware realism, then benchmarking, advanced modulation and system sizing, with validation as the eighth of nine tasks. By late July, the code contained formula errors that its checks had not caught: a transmit gain coefficient that left two missions 3.01 dB low, a pointing loss missing a factor of four, and a PPM error rate in which more background light made the link perform better. An early OPALS comparison had passed only because two of these errors cancelled. In order to reduce this form of error, every formula was then recorded with its source and equation number before it is coded, together with a validation suite that grew alongside the model. Whilst this cost time in the middle of the project, it is the reason the later results can be traced, checked and validated.

Three changes of scope followed from what the available sources and hardware allowed. The plan intended to benchmark against small-satellite missions, but CubeSat-class missions publish the least detail, so the benchmarks moved to flown missions with fuller published data, and the detailed DSOC paper became the strongest test. The stabilised laser could not be delivered to the Surrey Space Centre in time, so Objective 4 became an analytical study of coherent BPSK and DPSK, and the polarisation-based schemes named in the original plan were not modelled. The component costs planned for Objective 5 are not published in any available source and are recorded as unavailable. Additioanlly, two pieces of work outside the plan were added: the Surrey ground station study, prompted by a question from the supervisor, and the assessment of which benchmark comparisons carry independent weight, which only became necessary once the benchmarks were run.

In hindsight, running the measured comparisons earlier would have found errors sooner, as well as justified some of the design choices, leaving more time for writing. Some features were also built and later removed because the project did not need them, including a graphical interface and other ideas (as discussed in Chapter 2). A later reading of the validation suite, applying the evidence-versus-arithmetic test check by check (Section 4.3.5), found that about fourteen of the 65 checks could be passed by a trivial model or repeat another check; because the code was already frozen they are reported as run rather than removed, and the finding is recorded rather than acted on.

5.5 Future Work

Each extension follows from a limitation above. The validation suite itself should be pruned first: the checks identified in Section 4.3.5 that a trivial model also passes should be replaced by ones that cannot, for example testing the GOPEX seeing term against the night-to-night residual once the range term is removed, and the DSOC rate model against absolute flown rates rather than their ratio. The terminal register of Objective 5 should be coupled to the link model, so that the terminal class that closes a given link into Surrey, and the mass cost of the aperture-for-power trade of Section 4.6, can be sized directly. The Surrey results assume a clear, dark night, so the natural next step is a statistical availability model driven by Guildford cloud records, night length through the year and the elevation of a target, which would turn single-scenario rates into data volumes over a year; this is probably the most useful extension for the University. The station parameters could be measured directly: the optics throughput of the existing telescope, the seeing from the width of star images recorded with its camera, and the tracking precision of its mount. The model holds pointing error constant with range, and a range-dependent pointing budget would firm up the deep-space results. Adding solar conjunction geometry, for which coronal-turbulence models exist [63, 64], would allow the simulator to predict outage seasons rather than only clear-sky rates. Finally, the best test of the model rests on one power-limited mission. Each future power-limited flight, and any further published DSOC contacts, could be added to the suite to test the 0.7 dB threshold agreement more widely, or to show that it does not hold.

The most valuable extension is the one the project began with and had to shelve, the stabilised-laser hardware demonstration. Chapter 4 now specifies what that hardware must achieve, as a linewidth budget for each data rate, so the demonstration would start from a quantified target. Combined with a detector and mount survey for the 40 cm telescope, this would carry the work from simulation towards a working University of Surrey optical ground station.

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APPENDIX A: THE VALIDATION SUITE

Table A.1 lists every check in the automated validation suite of Section 3.7, generated directly from the suite so that it cannot drift from the code. Kind: reproduction of a published calculation, or comparison against a flight measurement. A check passes when $|\text{computed} - \text{expected}|$ is within the tolerance, in the quantity's own unit; the tolerances are discussed in Section 4.1. Basis gives how each tolerance was set: P, the printed precision of the published value, or the accuracy of reading it from a figure; S, the spread the source publishes for its measurement; J, judgement, where the source publishes no spread or the check tests closure or a trend. Sources are given in short form; the LLCD statement of "approximately one bit per photon" in checks 47 and 48 is from [65]. 65 of 65 pass: 48 reproductions and 17 measured comparisons; Section 4.3.5 discusses which of them a trivial model would also pass.

Table A.1: The validation suite: every check, its source, kind, computed and expected values, tolerance and its basis, unit and result.

#	Check	Source	Kind	Computed	Expected	Tol.	Basis	Unit	Result
1	g_T at $\alpha = 1.12, \gamma = 0$	Klein & Degnan 1974, Sec. II.B	reproduction	0.8145	0.8145	0.0005	P	–	pass
2	ISL free-space loss at 1000 km	Liang 2022, Table 2	reproduction	-258.2	-258.2	0.01	P	dB	pass
3	ISL required P_T at 1000 km	Liang 2022, Table 2	reproduction	15.32	15.32	0.02	P	dBm	pass
4	ISL free-space loss at 5000 km	Liang 2022, Table 2	reproduction	-272.2	-272.2	0.03	J	dB	pass
5	ISL required P_T at 5000 km	Liang 2022, Table 2	reproduction	29.3	29.3	0.02	P	dBm	pass
6	ISL free-space loss at 10000 km	Liang 2022, Table 2	reproduction	-278.2	-278.2	0.01	P	dB	pass
7	ISL required P_T at 10000 km	Liang 2022, Table 2	reproduction	35.32	35.32	0.02	P	dBm	pass
8	Slant distance at 10 deg	Liang 2022, Table 5	reproduction	1.81e+03	1.81e+03	0.1	P	km	pass
9	Mie loss at 10 deg	Liang 2022, Table 5	reproduction	-0.5715	-0.57	0.005	P	dB	pass
10	Geometric loss at 10 deg	Liang 2022, Table 5	reproduction	-1.216	-1.22	0.005	P	dB	pass
11	Up/downlink required P_T at 10 deg	Liang 2022, Table 5	reproduction	22.28	22.28	0.02	P	dBm	pass
12	Slant distance at 40 deg	Liang 2022, Table 5	reproduction	810.7	810.7	0.1	P	km	pass
13	Mie loss at 40 deg	Liang 2022, Table 5	reproduction	-0.1544	-0.15	0.005	P	dB	pass
14	Geometric loss at 40 deg	Liang 2022, Table 5	reproduction	-0.3284	-0.33	0.005	P	dB	pass
15	Up/downlink required P_T at 40 deg	Liang 2022, Table 5	reproduction	13.98	13.98	0.02	P	dBm	pass

Table A.1, continued

#	Check	Source	Kind	Computed	Expected	Tol.	Basis	Unit	Result
16	Slant distance at 90 deg	Liang 2022, Table 5	reproduction	549	549	0.1	P	km	pass
17	Mie loss at 90 deg	Liang 2022, Table 5	reproduction	-0.09924	-0.1	0.005	P	dB	pass
18	Geometric loss at 90 deg	Liang 2022, Table 5	reproduction	-0.2111	-0.21	0.005	P	dB	pass
19	Up/downlink required P_T at 90 deg	Liang 2022, Table 5	reproduction	10.42	10.42	0.02	P	dBm	pass
20	Mars TM transmit gain	Valentini 2021, Table III	reproduction	129	129	0.01	P	dB	pass
21	Mars TM total pointing loss	Valentini 2021, Table III	reproduction	-8.452	-8.45	0.01	P	dB	pass
22	Mars TM space loss	Valentini 2021, Table III	reproduction	-373.5	-373.5	0.05	P	dB	pass
23	Mars TM received power	Valentini 2021, Table III	reproduction	-131	-131	0.05	P	dBW	pass
24	Mars TM received photon flux	Valentini 2021, Table III	reproduction	-33.68	-33.67	0.05	P	dB(phe/ns)	pass
25	Mars TM link margin	Valentini 2021, Table III	reproduction	2.08	2.09	0.05	P	dB	pass
26	Optimum gain at $\theta_{\max} = 0.35\,\mu$rad	Valentini 2021, Eq. (17)	reproduction	129.1	129	0.15	P	dB	pass
27	P_{BGL}, 50.0 nW case	Giggenbach 2022, Table 2	reproduction	50.27	50	0.85	P	nW	pass
28	P_{BGL}, 88.0 nW case	Giggenbach 2022, Table 2	reproduction	87.96	88	1.23	P	nW	pass
29	P_{BGL}, 2.4 nW case	Giggenbach 2022, Table 2	reproduction	2.413	2.4	0.374	P	nW	pass
30	P_{BGL}, 4.2 nW case	Giggenbach 2022, Table 2	reproduction	4.222	4.2	0.392	P	nW	pass
31	HV5/7 C_n^2 at $h = 0$	Casanovas 2021, Eq. (47)	reproduction	1.727	1.727	0.0001	P	$10^{-14}\,\mathrm{m}^{-2/3}$	pass
32	HV5/7 Fried parameter at 0.5 μm, zenith	definitional (Fried 1966; Andrews & Phillips Ch. 12)	reproduction	4.96	5	0.1	P	cm	pass
33	SCPPM data rate M=4	Valentini 2021, Tables I-II	reproduction	517.3	517.3	5.173	P	kb/s	pass
34	SCPPM data rate M=64	Valentini 2021, Tables I-II	reproduction	97	97	0.97	P	kb/s	pass
35	SCPPM data rate M=256	Valentini 2021, Tables I-II	reproduction	32.33	32.33	0.3233	P	kb/s	pass
36	Fading loss at 1% outage, 40 cm, 10 deg	Moll et al. 2022, SPIE 12266 (15 passes, OSIRISv1)	measured	5.021	4.2	1.8	S	dB	pass
37	Fading loss at 1% outage, 40 cm, 20 deg	Moll et al. 2022, SPIE 12266 (15 passes, OSIRISv1)	measured	2.418	1.9	0.9	S	dB	pass

Table A.1, continued

#	Check	Source	Kind	Computed	Expected	Tol.	Basis	Unit	Result
38	Aperture averaging weakens toward horizon	Perlot et al. 2007, SPIE 6457 (KIODO, 847 nm, 40 cm)	measured	1	1	0.1	J	–	pass
39	Rytov variance at 1550 nm, Table Mountain	Birch et al. 2024, Opt. Express 32, Table 1 (zenith-corrected)	measured	0.06281	0.063	0.02	S	–	pass
40	DSOC 20.6 Mb/s decode threshold K_s	Biswas & Srinivasan 2026, Table III / Fig. 16	measured	-1.068	-0.3152	2	J	dB(counts)	pass
41	DSOC link budget, received power at Table III point	Biswas & Srinivasan 2026, Table III (design column)	reproduction	-118.3	-118.3	0.5	J	dBW	pass
42	DSOC signal counts per symbol in flight, Table III point	Biswas & Srinivasan 2026, Sec. III.F / Table III (K_s 0.93, 3σ 0.29)	measured	0.9184	0.93	0.29	S	counts	pass
43	DSOC beam divergence measured in flight	Biswas & Srinivasan 2026, Sec. III.E, Fig. 11: 13–15 μrad	measured	14.15	14	1	S	μrad	pass
44	DSOC rate ratio 0.56/2.68 AU (eff=0.25)	Biswas & Srinivasan 2026, Sec. III.F / Sec. I (133 vs 8.33 Mb/s, 2 W)	measured	12.62	12.03	2.5	J	dB	pass
45	DSOC rate ratio 0.56/2.68 AU (eff=0.3)	Biswas & Srinivasan 2026, Sec. III.F / Sec. I (133 vs 8.33 Mb/s, 2 W)	measured	13.59	12.03	2.5	J	dB	pass
46	DSOC rate ratio 0.56/2.68 AU (eff=0.38)	Biswas & Srinivasan 2026, Sec. III.F / Sec. I (133 vs 8.33 Mb/s, 2 W)	measured	11.38	12.03	2.5	J	dB	pass
47	LLCD 622 Mb/s link closes (margin in 0..15 dB)	Boroson 2009, Sec. III	measured	1	1	0.1	J	–	pass
48	LLCD photon efficiency at threshold	Boroson 2009: “approximately one bit per photon”	measured	2.143	0	4	J	dB(b/ph)	pass
49	BPSK homodyne photons/bit at 10^{-5}	Kikuchi 2016, after Eq. (60): 4.55	reproduction	4.547	4.55	0.01	P	–	pass
50	BPSK heterodyne photons/bit at 10^{-5}	Kikuchi 2016, after Eq. (74): 9.0	reproduction	9.095	9	0.1	P	–	pass

Table A.1, continued

#	Check	Source	Kind	Computed	Expected	Tol.	Basis	Unit	Result
51	DPSK bit rate needed by a 10 MHz laser pair	Kikuchi 1984, Sec. V-B: (1/T) $>$ 3 Gbit/s	reproduction	3.142	3	0.5	P	Gbit/s	pass
52	DPSK S/N for BER 10^{-9} at $\sigma_\phi = 0.20$ rad	Kikuchi 1984, Fig. 3 (digitised)	reproduction	16.03	16.1	0.5	P	dB	pass
53	GOPEX day 2 vs paper theory	Levine et al. TDA 42-114, Table 2 (expected photoelectrons)	reproduction	-6.232	-6.548	0.6	J	dB	pass
54	GOPEX day 2 vs flight measurement	Levine et al. TDA 42-114, Table 2 (mean from m_l, σ_N; Eq. 6)	measured	-6.232	-7.428	2.5	J	dB	pass
55	GOPEX day 3 vs paper theory	Levine et al. TDA 42-114, Table 2 (expected photoelectrons)	reproduction	-10.07	-9.655	0.6	J	dB	pass
56	GOPEX day 3 vs flight measurement	Levine et al. TDA 42-114, Table 2 (mean from m_l, σ_N; Eq. 6)	measured	-10.07	-9.693	2.5	J	dB	pass
57	GOPEX day 6 vs paper theory	Levine et al. TDA 42-114, Table 2 (expected photoelectrons)	reproduction	-12.47	-11.88	0.6	J	dB	pass
58	GOPEX day 6 vs flight measurement	Levine et al. TDA 42-114, Table 2 (mean from m_l, σ_N; Eq. 6)	measured	-12.47	-12.49	2.5	J	dB	pass
59	GOPEX day 7 vs paper theory	Levine et al. TDA 42-114, Table 2 (expected photoelectrons)	reproduction	-12.75	-13.09	0.6	J	dB	pass
60	GOPEX day 7 vs flight measurement	Levine et al. TDA 42-114, Table 2 (mean from m_l, σ_N; Eq. 6)	measured	-12.75	-10.59	2.5	J	dB	pass
61	GOPEX day 8 vs paper theory	Levine et al. TDA 42-114, Table 2 (expected photoelectrons)	reproduction	-14.43	-14.07	0.6	J	dB	pass
62	GOPEX day 8 vs flight measurement	Levine et al. TDA 42-114, Table 2 (mean from m_l, σ_N; Eq. 6)	measured	-14.43	-14.52	2.5	J	dB	pass

Table A.1, continued

#	Check	Source	Kind	Computed	Expected	Tol.	Basis	Unit	Result
63	NODE terminal mass total	Kingsbury 2015, Table 3.5	reproduction	550	550	0.5	P	g	pass
64	NODE terminal average power total	Kingsbury 2015, Table 3.6 (8.52 printed, 0.01 rounding)	reproduction	8.52	8.52	0.02	P	W	pass
65	PPM capacity at $K_b = 1$, worst K_s/M miss over 8 points	Hemmati 2011, Fig. 5 (digitised)	reproduction	0.169	0	0.3	P	dB	pass

APPENDIX B: SURREY STATION PARAMETERS

Table B.1 gives the full parameter set of the Surrey ODK16 station used for the Surrey results of Chapter 4, with the pessimistic and optimistic edges carried for each assumed value.

Table B.1: Surrey ODK16 station parameters used in the link model. Intervals give the pessimistic and optimistic edges of each assumed value.

Parameter	Value	Source
Wavelength	1550 nm	As the missions of Table 2.3
Aperture diameter	400 mm	Manufacturer specification [59]
Central obscuration	152 mm secondary (ratio 0.38)	Manufacturer specification [59]
Site	Guildford, 51.2° N, taken as sea level	–
Link distances	LEO (600 km), Moon, 0.1–2.68 AU	Table 4.6
Clear-air loss	Liang model at sea level, 40° elevation	[5]
Optics throughput	0.7 (0.5–0.85)	Assumed
Detector efficiency	0.5 (0.2–0.85), SNSPD-class	Assumed
Dark count rate	1000 s^{-1} (10^4–10^2)	Assumed for a single-pixel detector; DSOC's 64-pixel array measured 3.7–22 kcps in total [13]
Detector field of view	100 μrad (200–50)	Assumed
Filter bandwidth	1 nm (2–0.2)	Assumed
Site seeing	15 μrad (40–7.5)	Assumed; interval from GOPEX [34]
Receive pointing error	0 (ideal tracking)	Assumed
PPM orders; shortest slot	4–256; 0.5 ns	Assumed; the shortest slot DSOC flew [3]
Code gap to capacity	1.8 dB	DSOC measured [3]
Night sky background	None	Assumed; night radiance tabulated as zero [40]
Turbulence profile	Hufnagel–Valley 5/7, 1% outage	Standard profile [30, 37]
Operating elevation	40°	Chosen

APPENDIX C: SIMULATOR SOURCE CODE

The full source code of the simulator is available at:

`https://gitlab.surrey.ac.uk/ae01384/link-budget-simulator-for-deep-space`

This is the final version used in this thesis.

- `model/`: the simulator itself, covering the link budget, the atmosphere and turbulence models, detection and modulation, coding, the coherent and stabilised-laser models, and the terminal register.
- `validation.py`: the automated validation suite of 65 checks listed in Appendix A.
- `docs/equations.md`: the equation register, which gives the source, equation number and units of every formula used in the code.

The validation suite is run from the repository root with:

```
PYTHONPATH=model python validation.py
```